

PCB FAB & ASSEMBLY NOTES FOR DESIGNERS

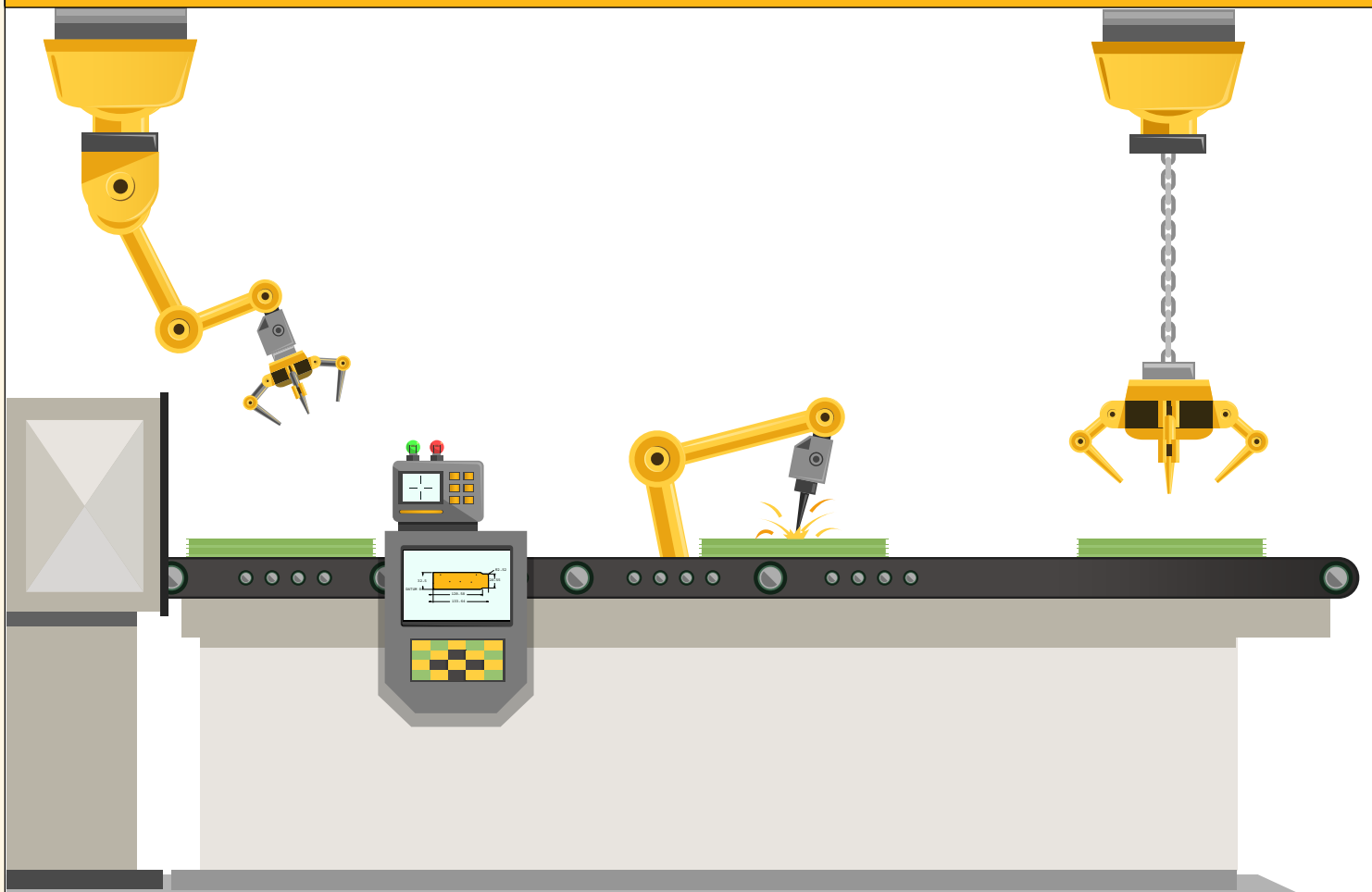


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1.1 Fab-friendly design starts with communication

Successful PCB manufacturing and assembly depends on clear, precise communication between designers, fabricators, and assemblers. Even a flawless design can fail if the manufacturing and assembly intent is not conveyed.

Fabrication houses rely on the documentation provided by designers, such as materials, stack-up configuration, and design intent, to manufacture and assemble the circuit board accurately.

Any ambiguity in this data can lead to delays, increased costs, or defective boards. To prevent such issues, your fab and assembly notes should convey the design requirements.

In this guide, you'll learn how to create practical fabrication notes, followed by best practices for writing clear and accurate assembly notes.

1.2 What are fabrication notes?

Fabrication notes, commonly known as fab notes, are detailed instructions from the PCB designer that communicate key manufacturing requirements. This data is an integral part of the documentation used by fabricators during production.

Typically included in the fabrication drawing, these notes specify requirements not captured in the Gerber/ODB++/IPC-2581 or CAD data. The details include material types, tolerances, surface finishes, compliance standards, and IPC classes.

Fab notes serve multiple purposes:

- Act as a checklist for manufacturing requirements.
- Reduce fabrication ambiguity and facilitates consistency in various production runs.
- Ensure the final product aligns with electrical, mechanical, and compliance expectations.
- Provide a reference point for all stakeholders and supports accurate quoting and procurement.
- Enable easier troubleshooting and revision tracking.

While modern PCB workflows increasingly rely on smart digital formats like IPC-2581 and ODB++ to streamline data exchange, fabrication notes in PDF documents still play a critical role.

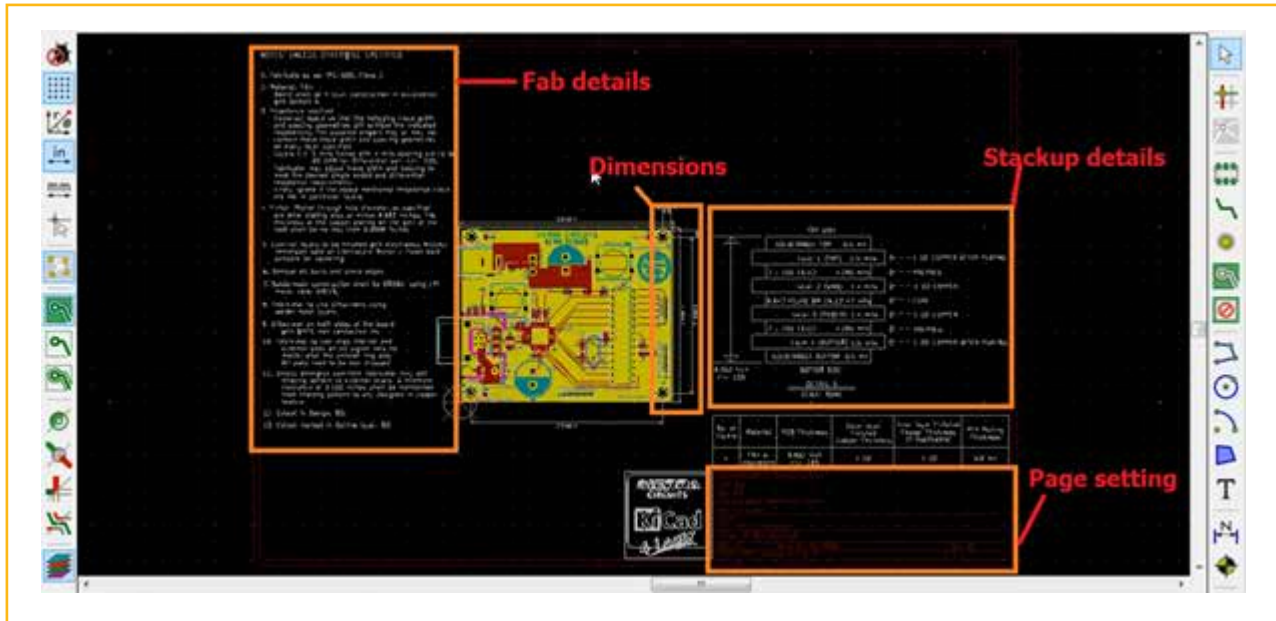
1.3 Role of fabrication drawings

Fabrication drawings are engineering documents that visually and dimensionally define the physical board. It also provides spatial and geometric context. While fab notes provide written specifications, the fabrication drawing acts as the graphical blueprint for manufacturing.

A complete fabrication drawing typically includes:

- Board outline with X-Y dimensions.
- Drill chart with hole sizes and quantities.
- Locations of plated and non-plated holes.
- Cut-outs, slots, and routing edges.
- Stack-up details, including material types and copper thicknesses.
- Fiducial locations and tooling holes.
- Manufacturing tolerances (general and specific).
- Desired surface finish type (e.g., ENIG, HASL).
- Solder mask and silkscreen details.
- Controlled impedance requirements (if applicable).

- Test point and test coupon locations.
- IPC class or relevant industry standards.
- Specific markings and labeling instructions.
- Panelization/array information (if applicable).
- Title block showing revision number, drawing date, and part number.



An example of a fab drawing

In some cases, additional views (e.g., cross-sections) are included to clarify multilayer construction, impedance requirements, or special instructions for flex or hybrid designs.

Fabrication drawings not only guide the manufacturing process but also serve as a contractual reference point. Manufacturers often quote and build boards strictly according to the drawing and associated notes. It is critical that the fabrication drawing and fab notes remain synchronized and are updated with every design revision.

1.4 Production files required for fabrication and assembly

- **Gerber (RS-274X):** It is the traditional industry standard for transferring PCB design data. They consist of separate files for each layer (copper, solder mask, silkscreen), drill data, and possibly outline data.

Name	Date modified	Type	Size
Pick Place	06-04-2024 01:45	File folder	
0000313298-B01-R1.RC5	06-04-2024 01:45	APR File	4 KB
0000313298-B01-R1.RC5	06-04-2024 01:45	EXTREP File	1 KB
0000313298-B01-R1.RC5.GBP	06-04-2024 01:45	GBP File	22 KB
0000313298-B01-R1.RC5.GTP	06-04-2024 01:45	GTP File	209 KB
0000313298-B01-R1.RC5	06-04-2024 01:45	REP File	2 KB
0000313298-B01-R1.RC5.RUL	06-04-2024 01:45	RUL File	1 KB
0000313298-B01-R1.RC5-assembly	06-04-2024 01:45	Adobe Acrobat D...	731 KB
0000313298-B01-R1.RC5-macro.APR_LIB	06-04-2024 01:45	APR_LIB File	25 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	APR File	14 KB
0000313298-C01-R1.RC5	06-04-2024 01:45	DRR File	3 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	EXTREP File	2 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	G1 File	411 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	G2 File	80 KB
0000313298-C01-R1.RC5.G3	06-04-2024 01:44	G3 File	411 KB
0000313298-C01-R1.RC5.G4	06-04-2024 01:44	G4 File	83 KB
0000313298-C01-R1.RC5.G5	06-04-2024 01:44	G5 File	435 KB
0000313298-C01-R1.RC5.G6	06-04-2024 01:44	G6 File	1,058 KB
0000313298-C01-R1.RC5.G7	06-04-2024 01:44	G7 File	1,064 KB
0000313298-C01-R1.RC5.G8	06-04-2024 01:44	G8 File	435 KB
0000313298-C01-R1.RC5.G9	06-04-2024 01:44	G9 File	75 KB
0000313298-C01-R1.RC5.G10	06-04-2024 01:44	G10 File	435 KB
0000313298-C01-R1.RC5.G11	06-04-2024 01:44	G11 File	56 KB
0000313298-C01-R1.RC5.G12	06-04-2024 01:44	G12 File	435 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	GBL File	243 KB
0000313298-C01-R1.RC5.GBO	06-04-2024 01:44	GBO File	9 KB
0000313298-C01-R1.RC5.GBS	06-04-2024 01:44	GBS File	64 KB
0000313298-C01-R1.RC5.GM	06-04-2024 01:44	GM File	1 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	GTL File	759 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	GTO File	1,975 KB
0000313298-C01-R1.RC5.GTS	06-04-2024 01:44	GTS File	341 KB
0000313298-C01-R1.RC5.LDP	06-04-2024 01:45	LDP File	1 KB
0000313298-C01-R1.RC5	06-04-2024 01:44	REP File	11 KB
0000313298-C01-R1.RC5.RUL	06-04-2024 01:44	RUL File	1 KB
0000313298-C01-R1.RC5	06-04-2024 01:45	Text Document	45 KB
0000313298-C01-R1.RC5-macro.APR_LIB	06-04-2024 01:44	APR_LIB File	52 KB

Gerber design data files are required during submission

- **ODB++:** This is an intelligent, single-file database format that contains all the necessary PCB design data, including layer information, drill data, netlist, component information, and stack-up details.

(ss ,p) sst	
(sp ,p) spt	
(sm ,p) smt	
(sig,p) top	
(pg ,p) gnd2	
(sig,p) sig3	
(pg ,p) gnd4	
(pg ,p) pwr4	
(pg ,p) pwr5	
(pg ,p) gnd7	
(sig,p) sig8	
(pg ,p) gnd9	
(sig,p) bottom	
(sm ,p) smb	
(sp ,p) sph	
(ss ,p) ssb	
(rt ,p) outline+1	

(drl,p) drill	
(doc,p) assy	
(doc,p) outline	
(doc,p) fab	
(doc,p) fab_drc	
(doc,p) height_top	
(doc,p) height_bot	

ODB++ design data files

- **IPC-2581:** The IPC-2581, also known as IPC-DPMX (digital product model exchange), provides a unified design data format that includes fab, assembly, and test data in a single file, enabling seamless data exchange between designers and fab/assembly houses.

L01_top	ART File	127 KB	No
L02_gnd	ART File	28 KB	No
L03_sig	ART File	65 KB	No
L04_gnd	ART File	37 KB	No
L05_pwr	ART File	35 KB	No
L06_pwr	ART File	38 KB	No
L07_gnd	ART File	28 KB	No
L08_sig	ART File	63 KB	No
L09_gnd	ART File	28 KB	No
L10_bot	ART File	77 KB	No
ncroutebits_auto	Text Document	1 KB	No
netlist	IPC File	69 KB	No
Outline	ART File	7 KB	No
PCB-0724_brassboard_main-1-10.r...	ROU File	1 KB	No
PCB-0724_brassboard_main-1-10	DRL File	14 KB	No
place_txt	Text Document	12 KB	No
pmh	ART File	12 KB	No
pmt	ART File	17 KB	No
smb	ART File	25 KB	No
smt	ART File	28 KB	No
ssb	ART File	92 KB	No
sst	ART File	207 KB	No

IPC-2581 design data submission files

IPC-2581 is recommended when available, as it integrates stack-up, netlist, BOM, and XY data in a single package, reducing the risk of misalignment or misinterpretation.

1.5 PCB fabrication files submission checklist

Here's the list of files you need to submit to the manufacturer to get your board fabricated.

File name/type	Description
Gerber (RS-274X)/ODB++/IPC-2581	Standardized file formats containing all PCB layer data (copper, solder mask, silkscreen, etc.) required for fabrication. ODB++ and IPC-2581 are advanced alternatives to traditional Gerber files.
NC drill (Excellon)	Specifies the locations, sizes, and types of all holes to be drilled in the PCB, including plated and non-plated holes.
Fab drawing	Outlines mechanical dimensions, board outline, stack-up, material specifications, and other critical fabrication notes.
Drill drawing	Shows the location, size, and type of all drilled holes, including via, mounting, and tooling holes, for manufacturing reference.
Stack-up file	Provides detailed information about the PCB layer structure, materials, and thicknesses.
Netlist (IPC-356A, IPC-D-350/356)	Industry-standard netlist file formats are used to verify electrical connectivity and test the fabricated PCB against the design intent.
Stencil file	Used to manufacture solder paste stencils, typically derived from the paste mask layers.
Testpoint reports / Test point file	Identifies test points for quality control and electrical verification, facilitating in-circuit and functional testing.
Schematic file	Circuit diagrams for reference, not always required for fabrication, but useful for troubleshooting.

What your manufacturer wants to see in your fab notes

02

Fabricators should get a detailed insight into your design's intent to produce boards accurately. Fab notes serve as the core communication channel for conveying essential technical design details that are not evident in Gerber files or layout data alone.

2.1 Stack-up details

The PCB stack-up defines the sequential arrangement of copper and dielectric layers. You need to clearly define the copper weights, dielectric thicknesses, and layer sequence. The thickness of individual copper-clad sheets should match the thickness present in the build-up diagram.

Include these details in your stack-up:

2.1.1 Material specification

Mention the materials used for each layer, including the core and prepreg. This involves detailing their specific types (e.g., FR-4, high-Tg, low-loss), dielectric constant (D_k), and dissipation factor (D_f) values at relevant frequencies.

Consider these guidelines when creating fab notes related to material specifications:

- Define copper foil type (e.g., electrodeposited, rolled annealed), surface treatment (e.g., RTF, VLP, HVLP), and thickness (e.g., 1 oz, 0.5 oz).
- Specify the resin properties.
- Indicate the glass fabric style and construction type, and provide cured thickness specifications with tolerances.

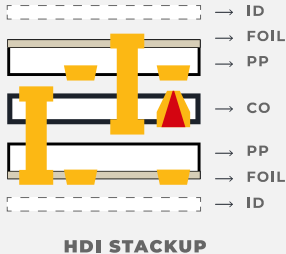
Instead of naming only commercial products (e.g., FR4, Rogers), use IPC-4101 slash sheet callouts wherever possible. Using IPC-4101 standards like /121, /124, or /126 is helpful because FR4 can vary widely among manufacturers.

These IPC standards define precise material properties (D_k , D_f , T_g), ensuring the manufacturer can select an exact-performing equivalent if needed.

Example note:

- Use woven E-glass/epoxy laminate and prepreg materials that comply with IPC-4101/126 or an approved equivalent.
- Alternative materials must be RoHS compliant, halogen-free, have a minimum glass transition temperature (T_g) of 170°C, and be approved by the company.
- Copper-clad sheet thicknesses should match those specified in the stack-up diagram.

Use Sierra Circuits' PCB Stackup Designer to create manufacturable build-ups quickly.



→ ID
→ FOIL
→ PP
→ CO
→ PP
→ FOIL
→ ID

HDI STACKUP

PCB DESIGN TOOL

Stackup Designer



TRY TOOL

2.1.2 Controlled impedance requirements

For designs with high-speed signals, controlled impedance is critical. Your fabrication notes must clearly outline all controlled impedance traces, specifying:

- Target impedance values (e.g., 50Ω single-ended, 100Ω differential), tolerance (e.g., ±5% or ±10%), and model (e.g., microstrip, stripline, differential pair).
- The corresponding trace widths and dielectric thicknesses are used to achieve these impedances.
- The layer on which the impedance traces exist.
- Indicate if a time domain reflectometry (TDR) test coupon is required for verification.

Example note:

- All 8 mil wide traces on outer layers must achieve 50Ω impedance.
- All 6 mil wide differential pairs on the inner layer must achieve 90Ω impedance.
- Vendors may adjust trace geometries by up to ±20% to meet impedance targets.

Sample impedance table:

The traces must adhere to the following trace width and spacing geometries. The supplied artwork may or may not contain these trace widths and spacing geometries on every layer specified.

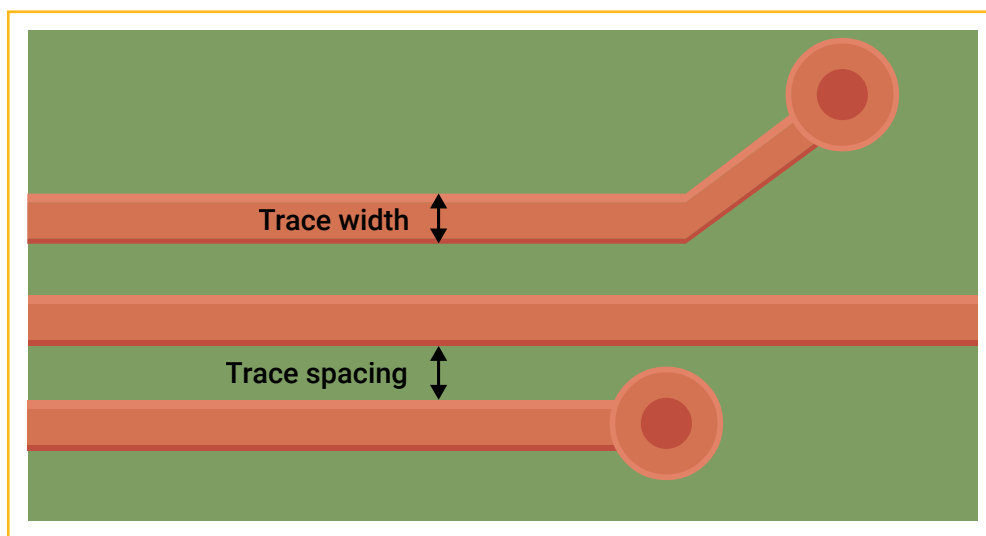
Layer	Type	Width/gap(mil)	Impedance	Tolerance
L1, L6	Microstrip	5.0	50Ω	±10%
L3	Differential pair	4.0/8.0	100Ω	±10%
L1	Coplanar	4.5/8.0	50Ω	±10%

2.1.3 Board/copper thickness and trace width considerations

Copper thickness affects current carrying capacity and thermal dissipation, while trace width impacts impedance and manufacturability. Providing these details helps fabricators optimize etching processes and layer construction to meet your electrical and mechanical design goals.

Specify:

- Final board thickness (e.g., 1.6 mm or 62 mil) and copper weight per layer (e.g., 1 oz, 0.5 oz).
- Minimum trace width and spacing.



Trace width and spacing on a PCB

- Controlled copper distribution for power planes.

Example note:

- Trace widths are to be measured at the base of the metallization.
- Minimum trace width requirements: 6 mil for outer layers and 4 mil for inner layers.
- Final trace widths and terminal areas must not deviate from the master pattern by more than ± 0.025 mm or 20%.

2.2 Drill sizes and tolerances

Precise drill specifications are vital for ensuring proper component fit (THT) and electrical connectivity (vias). Ensure the drill depth-to-diameter ratio complies with manufacturer capabilities (e.g., $\leq 10:1$ for standard processes).

Include a drill chart listing the following:

- Number of holes, drill size, hole tolerances
- Hole type (e.g., component hole, via, mounting hole, slot).
- Specify if holes are plated or non-plated. Use separate symbols for plated and non-plated holes to avoid misinterpretation.

DRILL CHART: TOP TO BOTTOM				
ALL UNITS ARE IN MILS				
FIGURE	FINISHED_SIZE	TOLERANCE_DRILL	PLATED	QTY
*	8.0	+2.0/-8.0	PLATED	692
*	10.0	+2.0/-10.0	PLATED	245
*	26.0	+3.0/-3.0	PLATED	8
*	28.0	+3.0/-3.0	PLATED	4
*	36.0	+3.0/-3.0	PLATED	4
*	40.0	+3.0/-3.0	PLATED	26
◆	91.0	+3.0/-3.0	PLATED	2
*	34.0X22.0	+3.0/-3.0	PLATED	2
*	45.0X20.0	+3.0/-3.0	PLATED	2
*	40.0	+2.0/-2.0	NON-PLATED	2
*	36.0	+2.0/-2.0	NON-PLATED	4

An example of a drill chart

Special drilling requirements:

- Mention the vias that require back drilling, the target depth, and the allowable tolerance.
- Microvia requirements for HDI constructions.
- Blind and buried via specifications.
- Sequential lamination requirements for complex via structures.

Example note:

- Ensure continuous electrolytic copper plating in all holes, with a minimum barrel thickness of 0.025 mm.
- Minimum finished hole size: 10 mil (after plating).
- Refer to the drill chart for finished hole sizes and tolerances.
- All holes must be positioned within 0.08 mm of their actual location as defined in the CAD data.

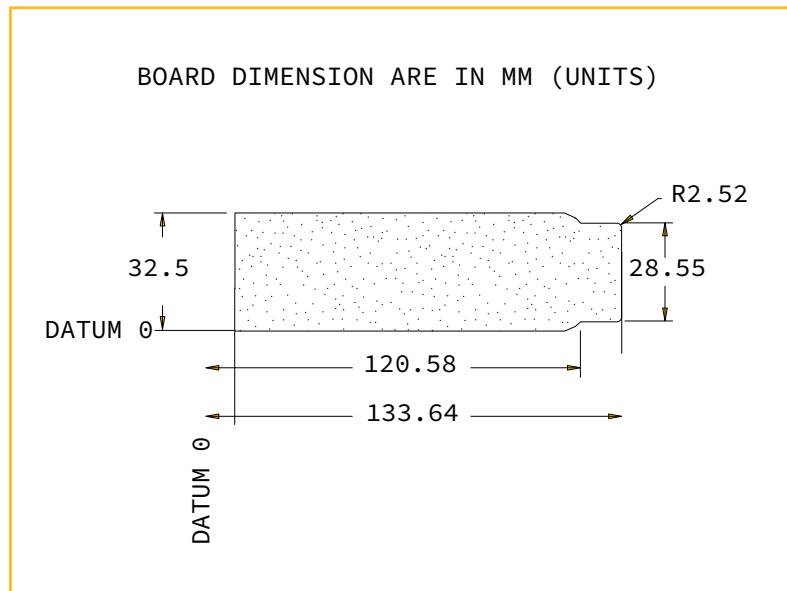
2.3 Board dimensions and cutouts

Precise board dimensions and details of any internal cutouts are essential for the accurate milling and shaping of the PCB.

Specify:

- Overall board dimensions in the X and Y axes with tolerances.
- Any internal cutouts, slots, non-rectangular outlines, or non-circular holes.
- Cutout sizes, tolerances, and locations relative to a common datum point.
- Edge finish specifications (routed, scored, or punched) and controlled depth cavities (if applicable).
- Tab and perforation requirements for panelization.
- Mounting hole locations and sizes.
- Tooling hole requirements for assembly.
- Fiducial placement and specifications.

Misalignment or inaccuracies in these dimensions can lead to assembly issues or fit problems in the final enclosure. Ensure the fab drawing includes outline dimensions with reference points.



A fab drawing specifying the board dimensions

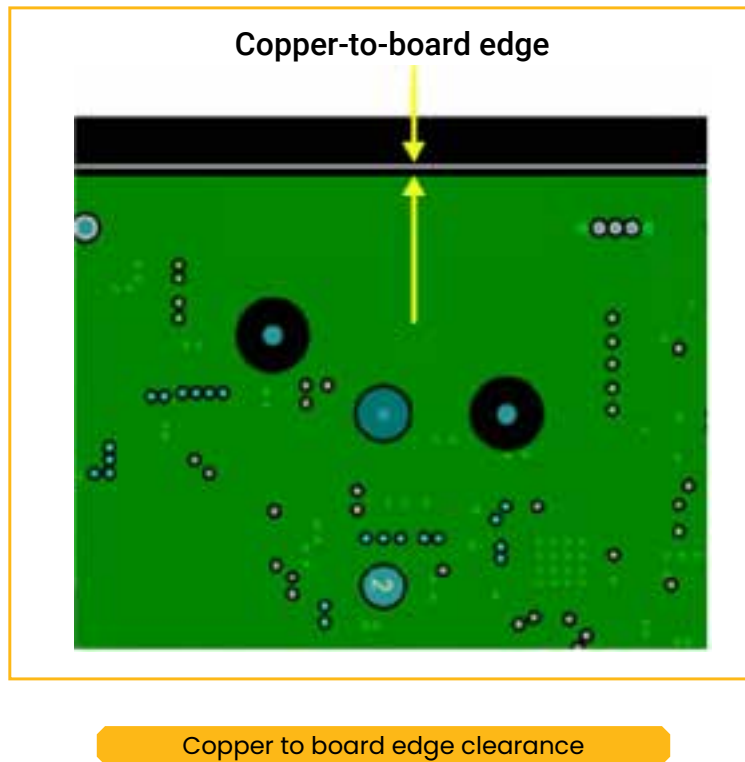
Example note:

- PCBs must be manufactured by the latest revision of IPC-6012, Class 2.
- All dimensions and tolerances should follow the current ASME Y14.5M standard.
- Drawings are not to be scaled.

2.4 Hole-to-edge and component clearance guidelines

To prevent manufacturing and assembly defects, you need to specify the following clearances clearly:

- Hole-to-edge clearance: Define the minimum distance required between any drilled hole (plated or non-plated) and the edge of the PCB.
- Copper-to-edge clearance: Specify the minimum distance between any copper feature and the board edge.

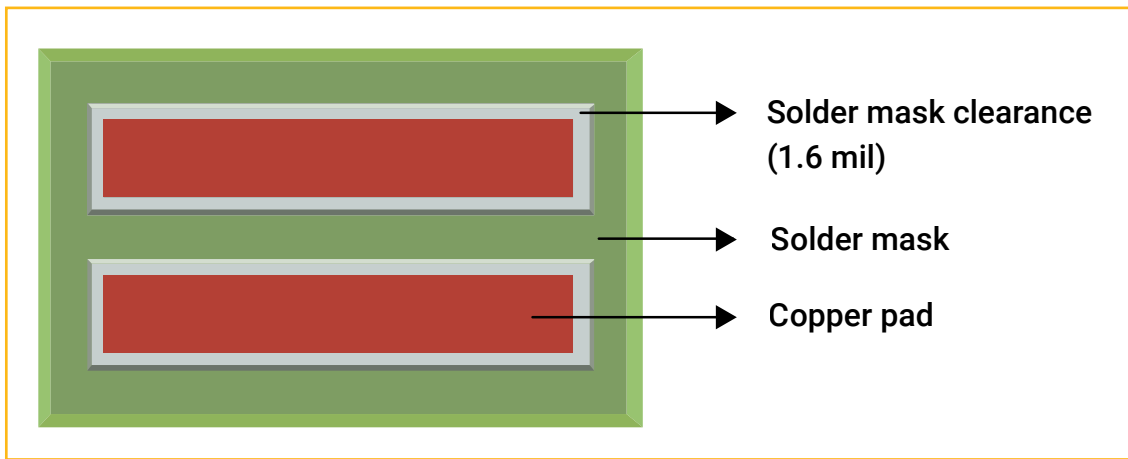


If your design has high-current connectors near the edge, add extra copper clearance to prevent damage during depanelization.

2.5 Solder mask and silkscreen specifications

The solder mask defines the areas where solder can be applied, while the silkscreen adds labels such as component names and reference designators for easy identification. Your fab notes for solder mask should specify:

- Solder mask type (liquid photo-imageable (LPI), dry film soldermask (DFSM), and epoxy liquid solder masks).
- Color specification (e.g., green, black, red).
- Standard solder mask thickness (typically 0.8 mil, specify if you have any other thickness requirements).
- Solder mask opening tolerances and registration.



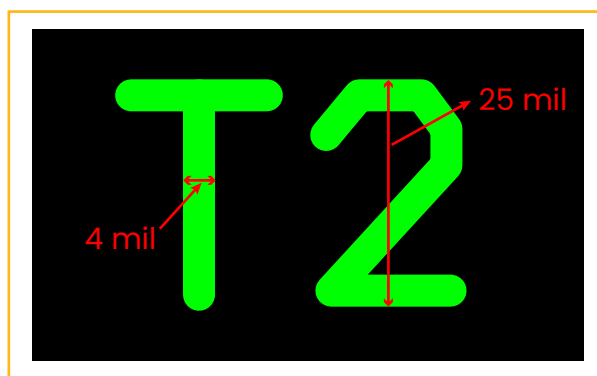
Solder mask clearance around pads

- Coverage (areas where solder mask should be removed, e.g., test points, gold fingers).
- Surface preparation and adhesion requirements.

Example note:

- Apply solder mask over bare copper (SMOBC) on both sides, using the artwork provided and following IPC-SM-840 Type B.
- The color should be matte green.
- Use liquid photo-imageable (LPI) solder mask, 0.001 – 0.002 mm thick, halogen-free.
- No solder mask bleed-out is allowed over exposed SMD pads, and no traces should be left exposed.

Silkscreen specifications:



Silkscreen layer on a PCB layout

- Ink type (non-conductive, permanent) and color requirements.
- Minimum line width and character size.
- Registration tolerances to board features.
- Vendor marking and UL marking locations.
- Pb-free or RoHS marking, if needed.
- Avoid silkscreen over pads or vias unless explicitly specified.

Example note:

- **Apply white, non-conductive, non-nutrient epoxy ink silkscreen on both sides.**
- **Unspecified stroke widths should be 0.13 mm.**
- **The silkscreen must be clipped away from exposed metal.**
- **Vendor date code, logo, UL, and other markings should be placed on the secondary side.**
- **For PCBs too small for marking, bag and tag labeling is acceptable.**

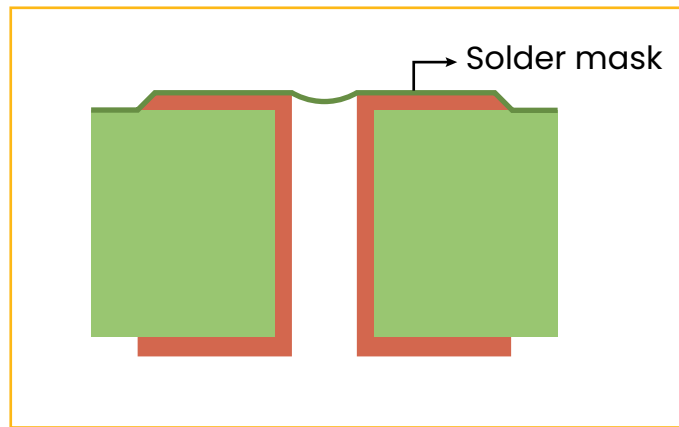
2.6. Via specifications: Tented, filled, or via-in-pad

Vias facilitate connections (electrically or thermally) between layers and require specific instructions for their treatment.

Specify how vias should be processed, such as tenting, filling, or using them in pad structures.

2.6.1 Via tenting

- Specify if vias should be tented (covered by solder mask) on one or both sides to prevent solder bridging or contamination.
- Define the required via tenting type (Type I(a), Type I(b), Type II(a), or Type II(b) by IPC-4761 standards). Provide the type if applicable.



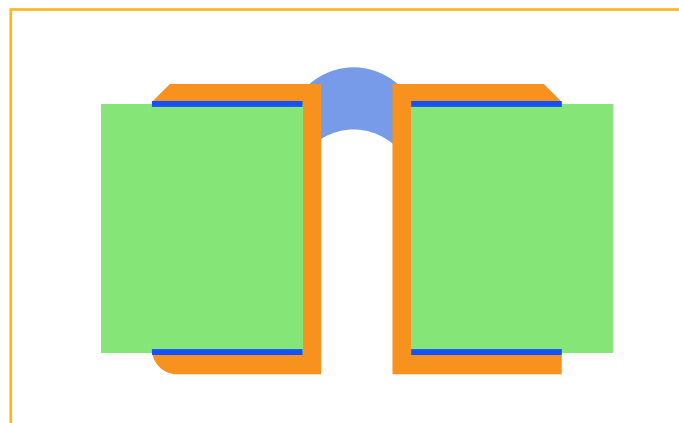
An illustration of a tented via

Example note:

- All vias on inner layers shall be tented with solder mask on both top and bottom sides.
- Create a Type II-b tented and covered structure as per IPC-4761.
- Tenting is required only on the top side for all non-plated through vias; specify Type I(a) as per IPC-4761.

2.6.2 Via plugging

- Specify if vias should be plugged to prevent solder ingress during assembly.
- Define the required via plugging type (Type III(a), Type III(b), Type IV(a), Type IV(b)).



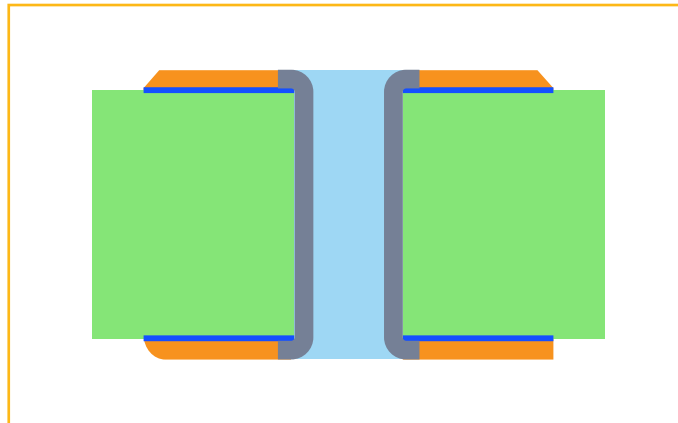
An illustration of a plugged via

Example note:

- Plug vias from the bottom side using non-conductive epoxy per Type III(a) of IPC-4761.
- All thermal vias under power components must be plugged with conductive epoxy per IPC-4761 Type IV(b).
- Ensure plugged vias (Type III(b)) are flush and planar to prevent stencil blockage during paste printing.

2.6.3 Via filling

- Always mention the type of via fill (conductive or non-conductive) and plating requirement.
- Define the required via filling type (Type V, Type VI(a), Type VI(b), or Type VII).



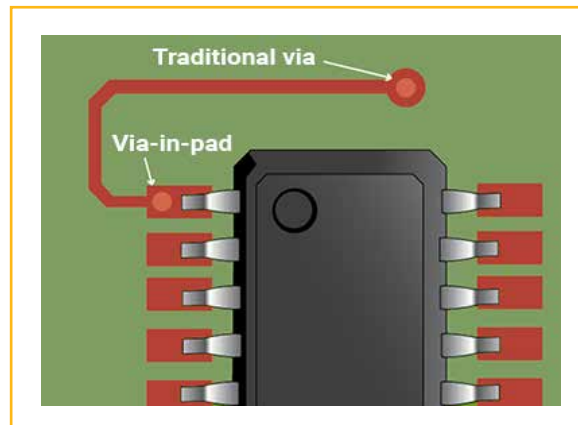
An illustration of a filled via

Example note:

- Fill all via holes with non-conductive epoxy and cap with copper plating as per IPC-4761 Type VI(a).
- Power and ground net vias to be filled with conductive epoxy and overplated per IPC-4761 Type V.
- All vias beneath component pads shall be filled and plated to ensure a flat surface for soldering.

2.6.4 Via-in-pad

For designs utilizing via-in-pad technology, explicitly state this requirement. These often require non-conductive fill and capping with copper or solder mask to create a flat surface for component soldering.



An illustration showing a via-in-pad and a traditional via

Example note:

- Via-in-pad is used under BGA and QFN components. Vias shall be filled with non-conductive epoxy, planarized, and capped with copper.
- All via-in-pad structures must comply with IPC-4761 Type VII. Ensure flush finish and no dimpling at the surface.
- Filled via-in-pad features to be copper capped and suitable for solder paste deposition; no mask tenting over pad.

2.7 Plating and surface finish requirements

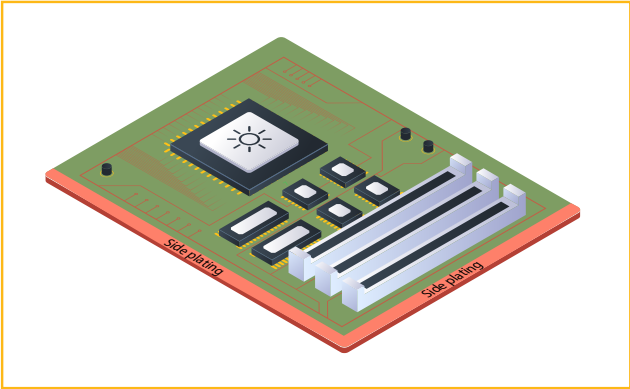
Surface finishes significantly impact solderability, shelf life, and electrical performance of your printed boards.

Consider these guidelines when writing fab notes for surface finishes:

- Clearly state the desired surface finish for the PCB pads and exposed copper areas. The table below lists the surface finish options based on your design requirement.

Surface finish	Suitability
HASL (hot air solder leveling)	Cost-effective, but not ideal for fine-pitch components
ENIG (electroless nickel immersion gold)	Flat finish; suitable for BGAs and RF circuits
ENEPIG (electroless nickel electroless palladium immersion gold)	Universal finish; compatible with both wire bonding and soldering
ImAg (immersion silver)	Suitable for high-speed boards; sensitive to humidity
ImSn (immersion tin)	Lead-free with flat finish; can be prone to tin whiskers

- Specify the desired gold and nickel thicknesses for finishes like ENIG.
- Mention if the finish applies to both sides of the board.
- If specific areas require different plating (e.g., hard gold for edge connectors), detail these requirements.
- If your design requires edge plating, make sure you include these details in your fab notes:



An illustration of an edge-plated PCB

- Specify which board edges require plating. To plate specific edges, provide a clear diagram with markings.
- If any planes need to connect to the side-plated area, add a note in the fab drawing specifying the layers.
- Mention the location of the side plating and the desired surface finish on a mechanical layer.
- If you're using jumpers, clearly specify their locations.

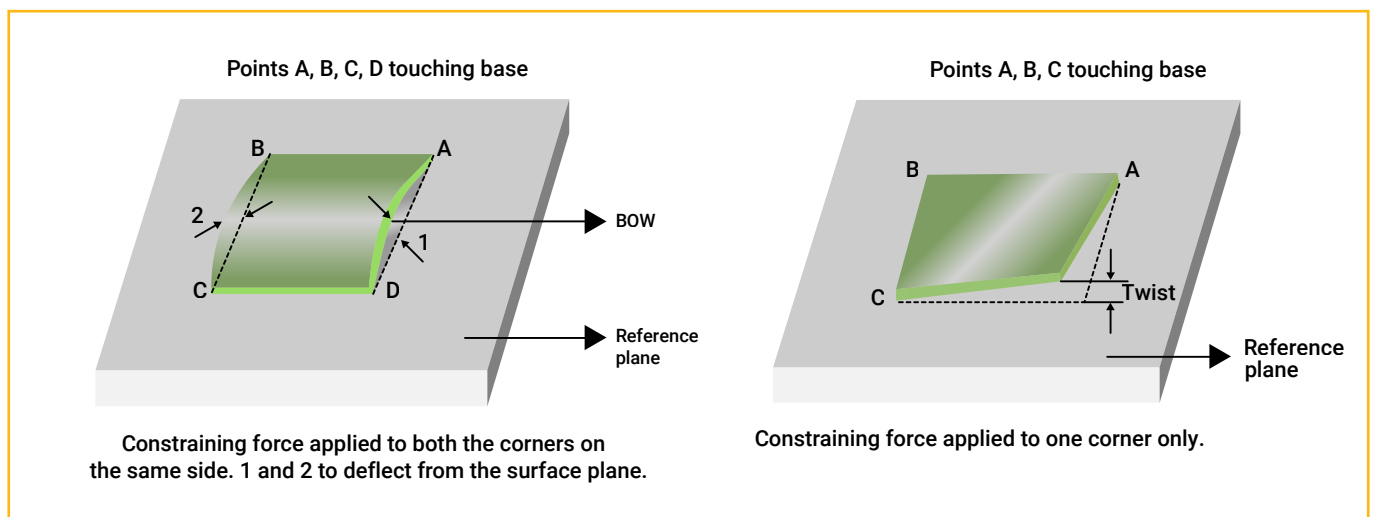
Example note:

- For ENEPIG finish, comply with the IPC-4556 standard.
 - Apply 118–236 microinches of electroless nickel, 2–6 microinches of electroless palladium, and 1.2 microinches of gold.
- For ENIG finish, follow the IPC-4552 standard.
 - Apply 118–236 microinches of electroless nickel and 2–5 microinches of gold.

2.8 Bow and twist tolerances

These specifications address the permissible deformation of the PCB after fabrication.

Define the maximum acceptable bow percentage (curve along the long axis) and twist (warping across the board) for the finished PCB.



Bow and twist observed on a PCB

Excessive warping can cause assembly issues and mechanical stress on components. Providing these tolerances enables fabricators to control manufacturing processes and deliver flat and stable boards.

Example note:

- The bow and twist of any sub-panel or individual PCB must not exceed 0.25 mm per mm length and should be tested by IPC-TM-650 2.4.22.

2.9 Manufacturing tolerances and inspection standards

Establish the overall quality level and inspection criteria for your board.

- State the general manufacturing tolerances that apply to all the board features (e.g., trace width/spacing, annular ring).
- Specify the target IPC performance class:
 - IPC class 2: Consumer electronics.
 - IPC class 3: High-reliability products (medical, aerospace, defense)

This dictates the overall quality and inspection standards.

- Mention the inspection standards (IPC-A-600 – acceptability of printed boards, IPC-6012 – qualification and performance specification) or methods that the manufacturer must adhere to.
- Define the IPC-6012 addendum based on the application:
 - IPC-6012 ES: Space and military applications .
 - IPC-6012 EM: Medtech devices.
 - IPC-6012 EA: Automotive industry.

Example note:

- **Fabricate as per IPC-6012 Class 2 standards. Dimensions and tolerances interpreted in accordance with ASME Y14.5M.**

2.10 Testing requirements

- Define electrical testing expectations:
 - Mention if 100% electrical testing (bare board E-test) is required
 - Name the required test methods (e.g., flying probe or bed-of-nails).
 - Additional testing requirements like Hi-Pot, continuity, or impedance verification
- Specify the purpose of the test coupons.
- Provide drawing instructions (dimensions, shape, and layout of the test coupons).
- Mention pad sizes, trace widths, and line spacing on coupons.
- State the number of test coupons and the intended locations on the PCB panel.

Example note:

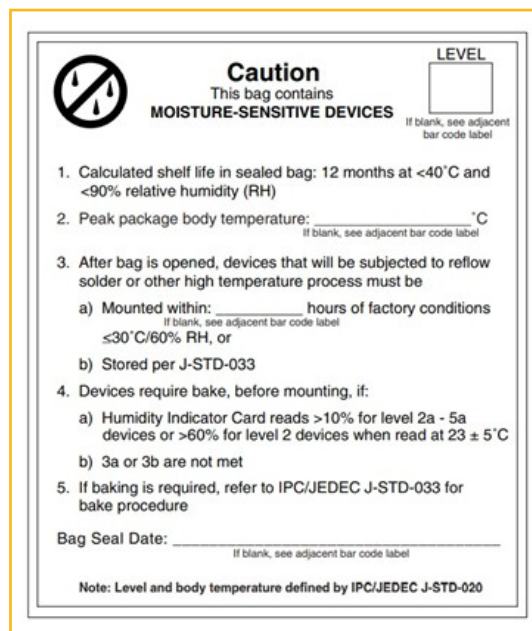
- All boards must pass a 100% electrical test using the supplied IPC-356 netlist, following the IPC-9252 standard.
- Include the micro-section test report with the shipment. Report to include layer thickness, annular ring, and plating thickness.
- A certificate of conformance must accompany each shipment.

2.11 Packaging and labeling instructions

Proper packaging and labeling ensure safe delivery and easy identification of the manufactured PCBs.

Specify:

- Packaging method (e.g., vacuum sealed, dry pack).
- Board quantity per pack or panel.
- Anti-static packaging for ESD-sensitive boards.
- MBB packaging, desiccant, and humidity indicator card (for moisture-sensitive builds) .



The image shows a template for a moisture-sensitive caution label. It features a 'Caution' header with a warning icon of a drop and a slash. Below the header, it states 'This bag contains MOISTURE-SENSITIVE DEVICES'. To the right, there is a 'LEVEL' box with a blank space for a bar code label. The label contains five numbered instructions regarding shelf life, temperature, and handling. At the bottom, there is a 'Bag Seal Date' field and a note referring to IPC/JEDEC J-STD-020.

Caution
This bag contains
MOISTURE-SENSITIVE DEVICES

LEVEL
If blank, see adjacent bar code label

1. Calculated shelf life in sealed bag: 12 months at <40°C and <90% relative humidity (RH)
2. Peak package body temperature: _____ °C
If blank, see adjacent bar code label
3. After bag is opened, devices that will be subjected to reflow solder or other high temperature process must be
 - a) Mounted within: _____ hours of factory conditions
If blank, see adjacent bar code label
≤30°C/60% RH, or
 - b) Stored per J-STD-033
4. Devices require bake, before mounting, if:
 - a) Humidity Indicator Card reads >10% for level 2a - 5a devices or >60% for level 2 devices when read at 23 ± 5°C
 - b) 3a or 3b are not met
5. If baking is required, refer to IPC/JEDEC J-STD-033 for bake procedure

Bag Seal Date: _____
If blank, see adjacent bar code label

Note: Level and body temperature defined by IPC/JEDEC J-STD-020

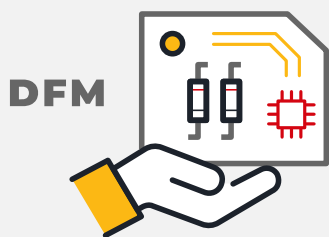
Moisture-sensitive caution label. Image credit: IPC

- Outer carton labeling: project name, part number, revision, date code.
- Include any specific shipping instructions or preferred carriers if applicable.
- X-out boards must be permanently marked and segregated.

Example note:

- Package PCBs in vacuum-sealed inner containers.
- Outer packaging must protect against damage during shipping and handling.
- Mark boards that do not meet all specifications with permanent markings on both sides.
- Package panels without X-outs together.
- Panels with up to 'n' X-outs should be packaged separately from those without X-outs.
- Reject panels with more than 'n' X-outs.

Use Sierra Circuits' Better DFM tool to check the manufacturability of your PCB design in one go.



PCB DESIGN TOOL

Better DFM



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Advanced fabrication notes and special requirements

03

Not all circuit boards follow standard construction. Applications that require higher performance, mixed technologies, or mechanical flexibility often demand advanced fabrication documentation.

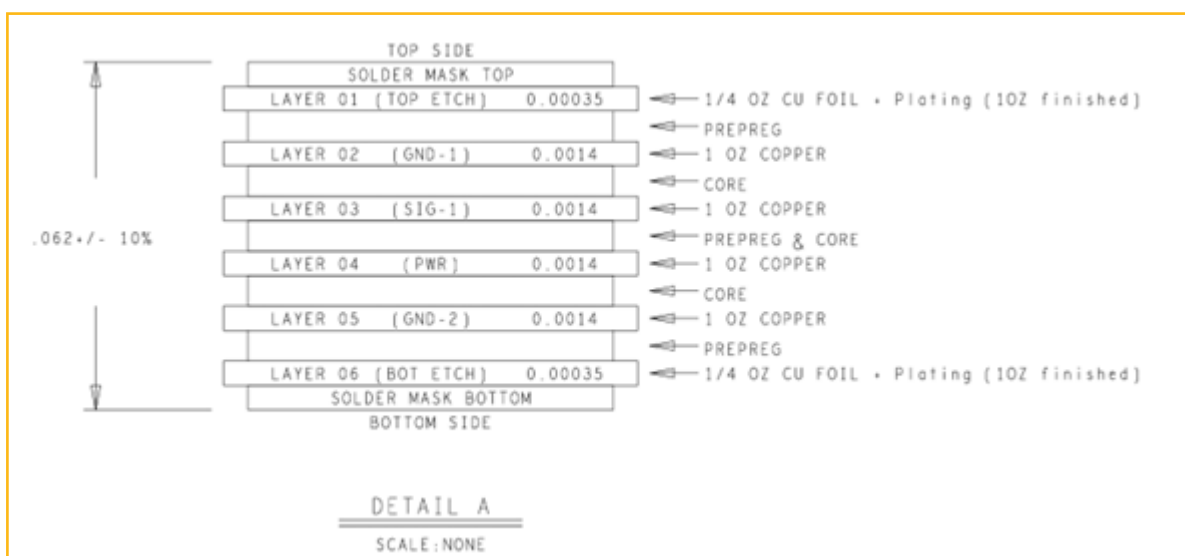
In such cases, conventional fab notes and drawings may not be sufficient to communicate the design intent. Here, you'll learn how to convey such requirements to your fab house.

3.1 Cross-section views in fabrication drawings

For complex stack-ups with blind/buried vias or unique material configurations, providing detailed cross-section views in the fabrication drawing is highly beneficial.

A cross-section view graphically illustrates the precise layer sequence, material types, and thicknesses, as well as the relationships between various drilled features (e.g., through-holes, blind vias, buried vias).

This visual aid eliminates ambiguity and helps the fabricator accurately interpret the design intent, particularly when multiple via types or non-standard build-ups are present.



An example of a cross-sectional stack-up drawing

Include cross-section diagrams when:

- The layer stack-up is non-standard or complex.
- Controlled impedance is required across multiple layers.
- There are blind, buried, or microvias.
- Different materials are used across layers (e.g., hybrid stack-ups).
- The design includes flex or rigid-flex sections.

Things to include in your cross-section drawing:

- Layer sequence: Signal, power, and ground layers in order.
- Material types: Core, prepreg, and special materials like Rogers or Isola.
- Thickness values: Dielectric and copper thickness per layer (e.g., 0.5 oz, 1 oz).
- Controlled impedance structures: Trace width, spacing, and reference planes.
- Via types and transitions: Blind, buried, microvia, and their impact on layer connectivity.
- Additional features: Solder mask, surface finish, and plating details.

Cross-section views should be prominently placed on the fabrication drawing, typically alongside the main board outline and stack-up table, to ensure immediate visibility and reference.

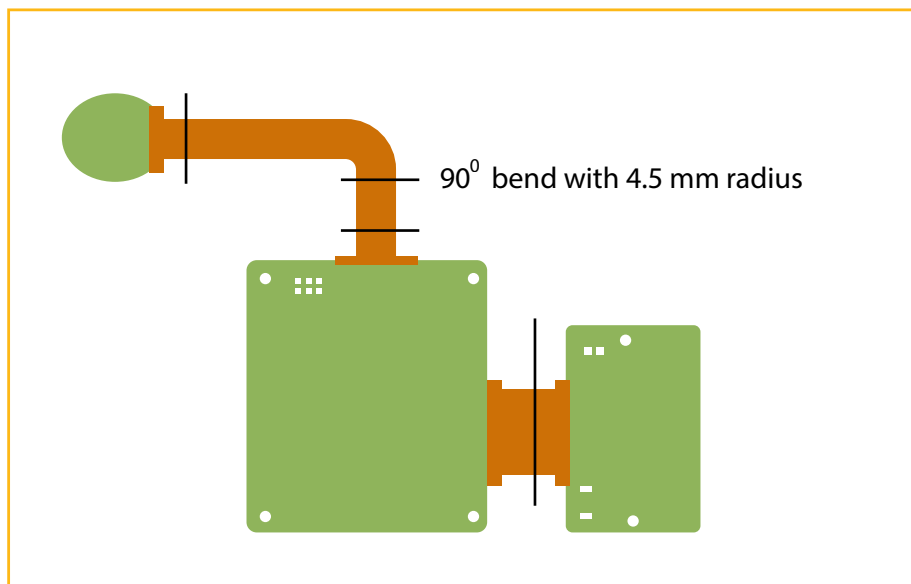
3.2 Flex and rigid-flex fab notes

Flexible and rigid-flex PCBs require highly detailed fabrication notes. These boards call for unique details, such as bending radius, adhesive layers, stiffeners, and dynamic mechanical stress during operation. Without clear instructions, even minor oversights can lead to reliability issues or early failure.

Key elements to include in flex/rigid-flex fabrication notes:

- State IPC standards (IPC-6013) and class (2/3).

- Specify the dielectric materials (e.g., polyimide film, flexible adhesive) and their thicknesses.
- Mention the stiffener requirements (e.g., polyimide, FR-4), including their material, thickness, and location.
- Define any areas designated as flex zones and specify the minimum allowable bend radius for the flexible sections. This is critical for preventing material fatigue and ensuring long-term reliability. Example: Minimum bend radius: 10x flex thickness.



Defining rigid and flex sections

- Detail the type and thickness of coverlay material. Specify any areas where coverlay should be absent (coverlay-less design) and indicate the required surface finish for these exposed flex circuits.
- Provide notes on adhesive flow control (e.g., use low-flow or no-flow prepreg in rigid-flex transitions to prevent resin bleed) to avoid adhesive bleed into flex areas, which can compromise flexibility.
- Set tighter tolerances for outlines due to the nature of flexible materials compared to rigid boards. Include any specific tool requirements for profiling.
- Include layer transitions, rigid-to-flex interfaces, and any z-axis constraints.
- Mention the number of flexing cycles expected during operation.

- Specify any misregistration between layer, if applicable. Misregistration shall not exceed ± 0.005 .
- Use S-parameter methods for impedance testing in hatched plane regions instead of TDR.
- State ZIF connector specs and location, if any.

Example note:

- This board shall be built to IPC-6013 Class 3 standards. Refer to the fabrication drawing for rigid and flex region definitions and full layer stack-up details.
- Flex zones must maintain a minimum bend radius of 10x the total flex thickness to prevent material fatigue during operation.
- Apply 25 μm polyimide coverlay over all flex areas unless otherwise noted. For coverlay-less zones, use ENIG surface finish and maintain 0.15 mm solder mask clearance.
- Apply 0.8 mm FR-4 stiffeners at connector regions as shown in the fab drawing. Bond using high-temp adhesive with minimum peel strength of 7 N/cm.
- Use no-flow prepreg at rigid-flex interfaces to prevent adhesive bleed into flex areas. Maintain 0.5 mm clearance between resin and flex outline.

3.3 Hybrid stack-ups and material combinations

Hybrid stack-ups incorporate two or more types of laminate materials within the same board. This is typically done to balance electrical performance (e.g., low-loss RF layers) with cost-effective construction (e.g., standard FR4 for digital sections).

- Indicate the regions of the board where different materials are used. This typically applies to designs combining standard FR-4 with high-frequency laminates (e.g., Rogers, Isola) in specific areas or layers. Dk and Df values should be identified for RF layers.

Imp	Lyr	Type	Image	Thk (Mil)	Name	Family
	sm1			0.5	~ ~	
3xΩ	I1s	Mixed		10.7	I-Speed 0.0107 H1xHH CA 18...	I-Speed
	I2p	Powe...				
				6.15	FR-370HR 1080 18x24 RC71	FR-370HR
				6.15	FR-370HR 1080 18x24 RC71	FR-370HR
1xΩ?V	I3p	Mixed		21	FR-370HR 0.021 HHxHH CM 18x12	FR-370HR
	I4p	Mixed				
				6.15	FR-370HR 1080 18x24 RC71	FR-370HR
				6.15	FR-370HR 1080 18x24 RC71	FR-370HR
3xΩ	I5p	Powe...		10.7	I-Speed 0.0107 H1xHH CA 18...	I-Speed
	I6s	Mixed				
	sm6			0.5	~ ~	

Sample hybrid stack-up of Rogers 4350-LoPro and Isola 370HR with blind and buried vias

- Specify the bonding materials and processes required to bond dissimilar laminates effectively, ensuring good adhesion and thermal management. Prepreg resin systems must be matched across cores for proper bonding and lamination.
- Mention the impedance value, targeted layer, and allowed tolerance for controlled impedance compliance. The manufacturing tolerance for stack-up includes layer-to-layer registration, board thickness, bow, and twist.
- Verify if the stack-up aligns with your design by requesting a cross-section analysis. This analysis will include details such as the dielectric thickness of each layer, copper plating on the outer layers, copper thickness on the inner layers, hole diameter, and plating.
- Detail how the interface between different material types should be managed, particularly concerning drilling and plating processes, as materials with varying thermal expansion coefficients can pose challenges.

Example note:

- **This design uses a hybrid stack-up with Rogers R04350B for RF signal layers (L2-L3) and standard FR-4 for remaining layers. Clearly define material zones as per stack-up drawing.**

- Use compatible low- D_k prepreg (e.g., 4450F) between dissimilar materials. Ensure bonding resin matches core materials to prevent delamination or resin starvation during lamination.

3.4 Embedded components integration

Embedded PCB components are passive devices fabricated within the substrate using conductive and dielectric materials. To integrate embedded parts in your layout, you must create an additional layer and must cover the following details:

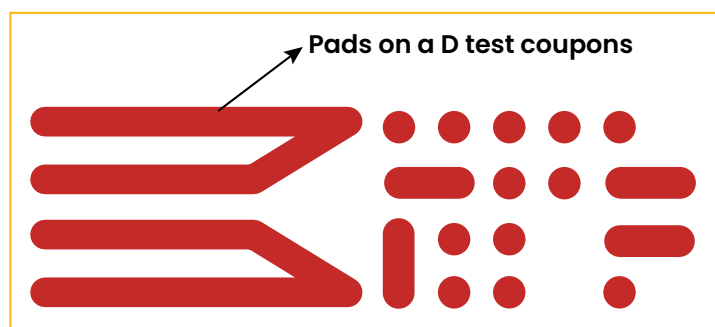
- Indicate the layer and orientation of each embedded component.
- Add XY coordinates and component specifications.
- Provide precise dimensions and tolerances of cavities.
- Specify the depth of each cavity accurately.
- Define the shape of each cavity and any special features.
- Add the minimum clearance between the cavity walls and the embedded component to allow for assembly and manufacturing variations.
- Attach datasheets for all embedded components, including specifications, dimensions, and thermal characteristics.

Example note:

- Embedded resistor located on layer L3 between coordinates X50.0, Y25.0 and X60.0, Y25.0.

3.5 Special fabrication requirements

- **D coupon testing:** Request D coupon samples for cross-section analysis of plating, layer registration, and other critical parameters. Specify the number of coupons required and their location on the panel.



Pads on a D coupon

Example note:

Provide D-coupon test samples on both ends of the production panel.

Minimum: 2 coupons per panel. Coupons to be used for layer registration, copper plating, and dielectric verification.

- **IST (interconnect stress test):** If reliability testing is required, indicate that IST must be performed to assess the durability of plated through-holes and vias under thermal cycling. Provide details on the test method and acceptance criteria.

Example note:

Perform IST testing per IPC-TM-650 Method 2.6.26 to evaluate PTH reliability. Test 3 sample boards per lot. Acceptable: $\leq 5\%$ resistance change post thermal cycling.

- **Hole castellation:** If castellated holes (half-plated holes on the board edge) are needed for board-to-board connections, define their location, size, and plating requirements.



Castellated circuit board

Example note:

Add castellated holes ($\varnothing$ 0.8 mm) with 50% plated arc on board edges as shown in fabrication drawing. Minimum copper thickness: 25 μ m.

- **Cavities and dimensions:** For boards with recessed cavities, clearly indicate the cavity location, depth, and dimensions. Specify any special tolerances or surface finish requirements for the cavity area.

Example note:

Machine cavity in top layer, 1.0 mm deep, as shown in layer cut-out detail. Tolerance: ± 0.05 mm. Maintain $R_a \leq 1.6$ μ m surface finish inside cavity.

- **Countersinks:** If countersink or counterbored holes are required, provide detailed specifications including diameter, depth, angle, and location. Reference the appropriate mechanical drawing or section view.

Example note:

Apply 90° countersink at specified holes on the bottom side. Final diameter: 4.0 mm; depth: 2.0 mm. Refer to mechanical detail M3 in fab drawing.

- **Wire bonding pads or special pads:** For wire bonding or other special pad requirements, specify which pads must remain free of solder paste or surface finish. Indicate these pads in the fabrication and assembly drawings, and any special cleaning or preparation steps to be followed.

Example note:

Wire bonding pads on Layer 1 shall remain free of ENIG. Do not apply a solder mask or paste. Clean using ionic rinse and mark as 'No Finish' in fab and assembly drawings.

Introduction to PCB assembly documentation

Once a PCB has been fabricated, the next step is assembly, soldering components on the bare board. While fabrication notes guide how the board should be built, assembly documentation ensures that components are placed accurately, soldered reliably, and appropriately inspected.

4.1 What are assembly notes?

Assembly notes are written instructions provided by the PCB designer or design team to communicate how components should be mounted and soldered on the board.

They complement the assembly drawings and ensure that the CM (contract manufacturer) understands the assembly requirements. These notes are handy for addressing components with unique placement rules, clarifying part substitutions (DNI/DNP), or describing the exceptions not reflected in the pick-and-place files.

4.2 Differences between fabrication and assembly notes

Though both are forms of technical communication, fabrication notes and assembly notes serve different purposes:

Aspects	Fabrication notes	Assembly notes
Purpose	Define how the bare board is manufactured	Define how components are placed and soldered
Audience	Fab house	Assembly shop
Information enclosed	Materials, stack-up, drills, tolerances	Component mounting and soldering instructions, polarity, orientation marks, pin 1 marking
Related files	Gerber, IPC-2581, ODB++, fabrication drawing	BOM, XY files, assembly drawing
Standards reference	IPC-6012, IPC-4101, IPC-TM-650	J-STD-001, IPC-A-610, IPC-1066

Keep fabrication and assembly notes in separate sections or documents to reduce confusion and ensure they are read by the right team.

4.3 Key documents a PCB designer should provide for assembly

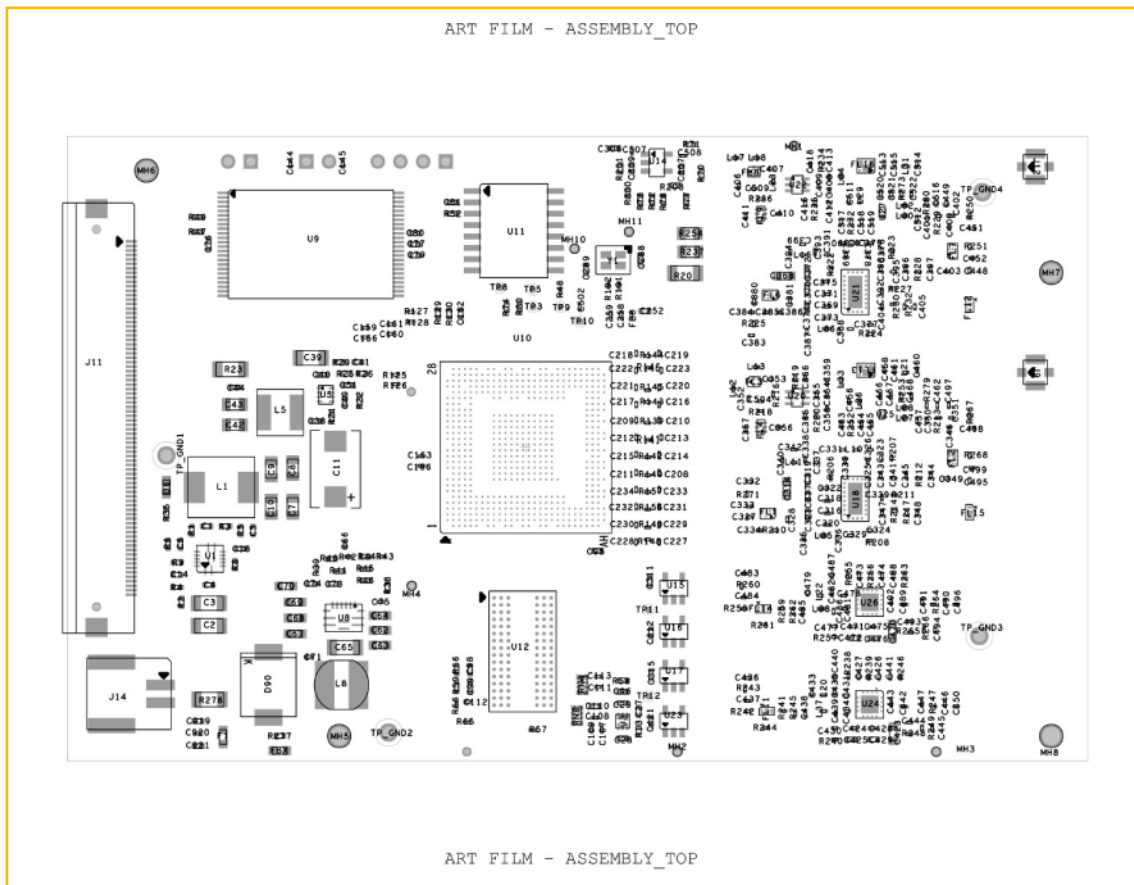
To ensure an accurate assembly process, you must compile and submit a comprehensive set of documents. These files give the assembly house all the necessary information to source components, program automated assembly equipment, and perform quality checks.

4.3.1 Assembly drawings

Assembly drawings depict the placement, orientation, and reference designators of all components. These drawings serve as the primary visual guide for manual assembly operations and quality control. They include:

- Reference designators (e.g., R1, U3, C5).
- Top/bottom/side views of the board .
- Component outlines, shapes, and locations.
- Orientation markers (e.g., pin 1 indicators).
- Designation of top and bottom sides.
- Mechanical details such as keep-out zones, tooling holes, and height restrictions.

Identify the primary (top) side. If components exist on both sides, provide separate drawings for each. Include projection symbols in the title block to avoid confusion.



An example of an assembly drawing

4.3.2 Bill of materials (BOM)

The BOM is a list of all components to be placed on your circuit board. This file includes part numbers, descriptions, quantities, and reference designators.

A well-organized BOM ensures that the correct parts are used throughout the assembly process, reduces sourcing errors, and simplifies quoting and assembly.

For a detailed explanation of BOM, refer to chapter 6.

4.3.3 Pick-and-place files (X-Y data)

These files provide the precise placement location and rotation angle of each surface-mount device on the board. Pick-and-place files, also known as centroid files, are essential for programming automated assembly machines, ensuring accurate component positioning. They typically include:

- X and Y coordinates.
- Rotation (in degrees).
- Reference designators.
- Layer information (top/bottom).
- Component packages (e.g., 0603, QFN).

4.3.4 Netlist

A netlist is a simple text file that shows how all the components on a PCB are connected. It details each component's pins and specifies which other pins they should connect to.

It is used for:

- Electrical testing (e.g., flying probe, ICT).
- Verifying continuity during debugging.
- Detecting open or short circuits after assembly.

Submit the netlist in a standard format (e.g., IPC-D-356 or as part of IPC-2581) to ease testing and troubleshooting procedures.

4.3.5 PCB assembly data submission checklist

Make sure you submit these files if you opt for assembly services.

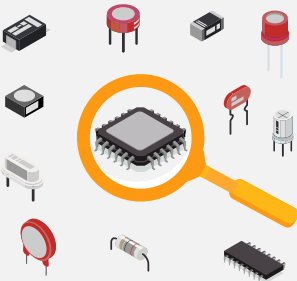
File names	Description
BOM	Comprehensive list of all components, including part numbers, values, manufacturers, and reference designators for procurement and assembly.
Pick and place/ Centroid files	Machine-readable data detailing the exact X/Y coordinates, rotation, and side (top/bottom) for automated placement of surface-mount components.
Component placement list	List of all component locations and orientations, useful for manual assembly or inspection.

*Cont'd.

File names	Description
Top/Bottom assembly	Files or drawings indicating the placement and orientation of all components on the top and bottom sides of the PCB for assembly reference.
APR Library	Assembly process requirements library, providing detailed information on component placement, orientation, and special assembly instructions.
EXTREP	External report files, often used to communicate additional manufacturing or test requirements not covered in standard files.

By submitting a complete and well-formatted assembly documentation package, you enable your CM to prepare accurate tooling, avoid placement errors, and assemble your board to specification without delays.

Use Sierra Circuits' Component Selector to choose the right electronic parts for your PCBs.



PCB DESIGN TOOL

Component Selector



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What your assembly technicians want to see in your assembly notes

05

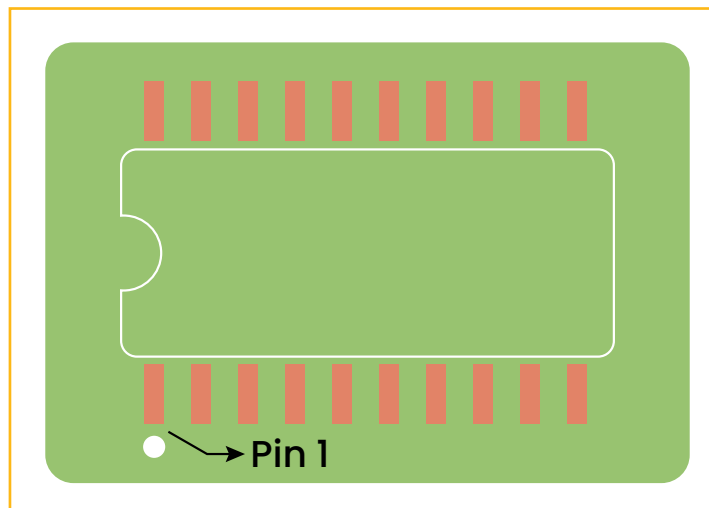
In addition to the standard assembly documents, clear and comprehensive assembly notes are crucial for an error-free assembly process.

Well-written assembly notes offer guidance on component orientation, soldering techniques, inspection criteria, and any special handling or post-assembly steps. They help eliminate guesswork, support compliance, and ensure the final product meets your functional and reliability requirements.

5.1 Component orientation and polarity

Incorrect component orientation is one of the most common causes of field failures and rework. Your assembly notes should clearly state:

- The orientation (e.g., pin 1 is marked with a dot on the silkscreen)



Pin 1 marking for an IC

- Any exceptions where components are rotated from standard orientations
- How to handle polarized components such as:

- Diodes (cathode marking).
- Electrolytic capacitors (positive/negative side).
- ICs and connectors (pin 1 location).

Reinforce this visually in the assembly drawing. If space is limited on the silkscreen, provide an external legend or separate orientation diagram.

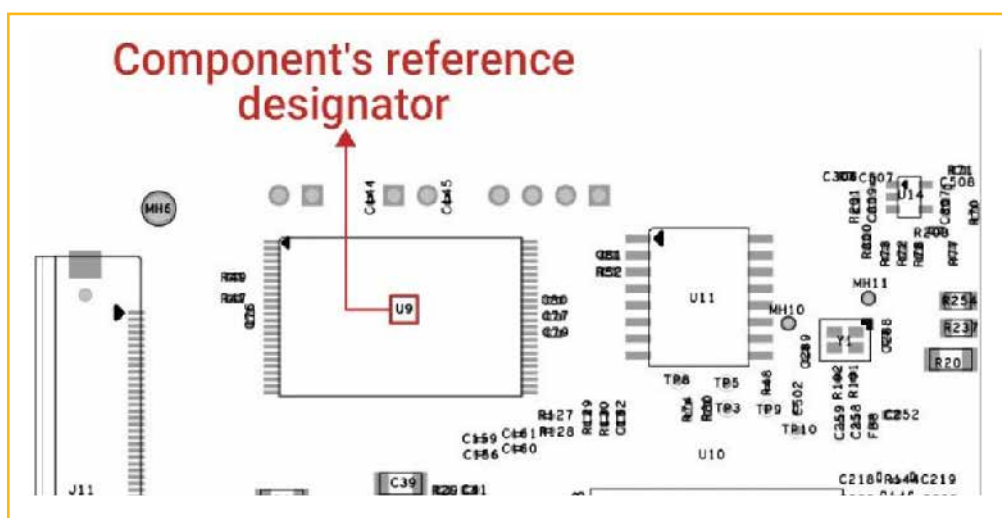
Example note:

- All polarized capacitors must be oriented per the silkscreen markings.
- Silkscreen marking indicates a positive polarity pad for polarized capacitors and a cathode polarity pad for diodes and LEDs.
- Diode D1 must be placed with the cathode end aligned with the white stripe.

5.2 Reference designators

Reference designators (e.g., R1, C4, U2) are the labels used to identify components during assembly, inspection, and testing. Your assembly notes should:

- Ensure that all designators in the BOM match the ones in the layout and drawing.



An example of an assembly drawing

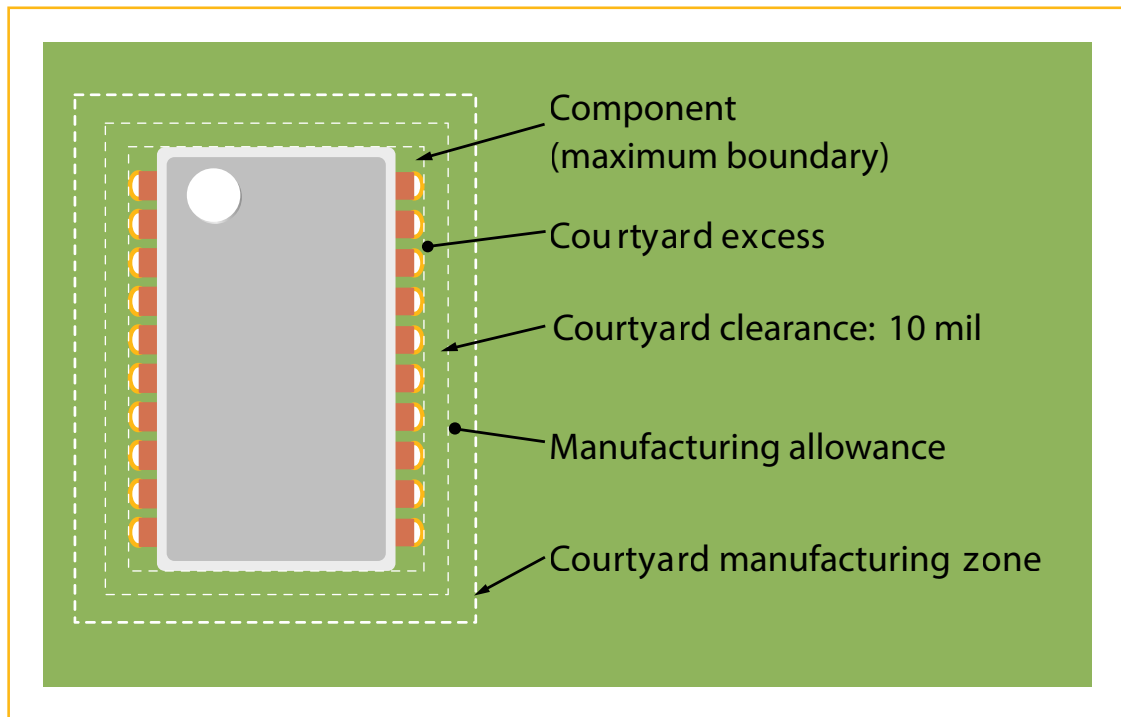
- Specify that designators must be visible on the board after assembly, wherever possible. If a component obscures the reference designator or needs to be relocated due to space constraints, include notes to help the assembler identify the part.
- If silkscreen markings (like reference designators) are omitted due to limited board space, be sure to provide alternative identification instructions. This can include referencing the bill of materials and using clearly labeled component outlines on the assembly drawing to ensure accurate placement during assembly.

Example note:

Reference designators shall be printed inside component outlines where feasible. If silkscreen coverage is not possible, refer to assembly drawings for identification.

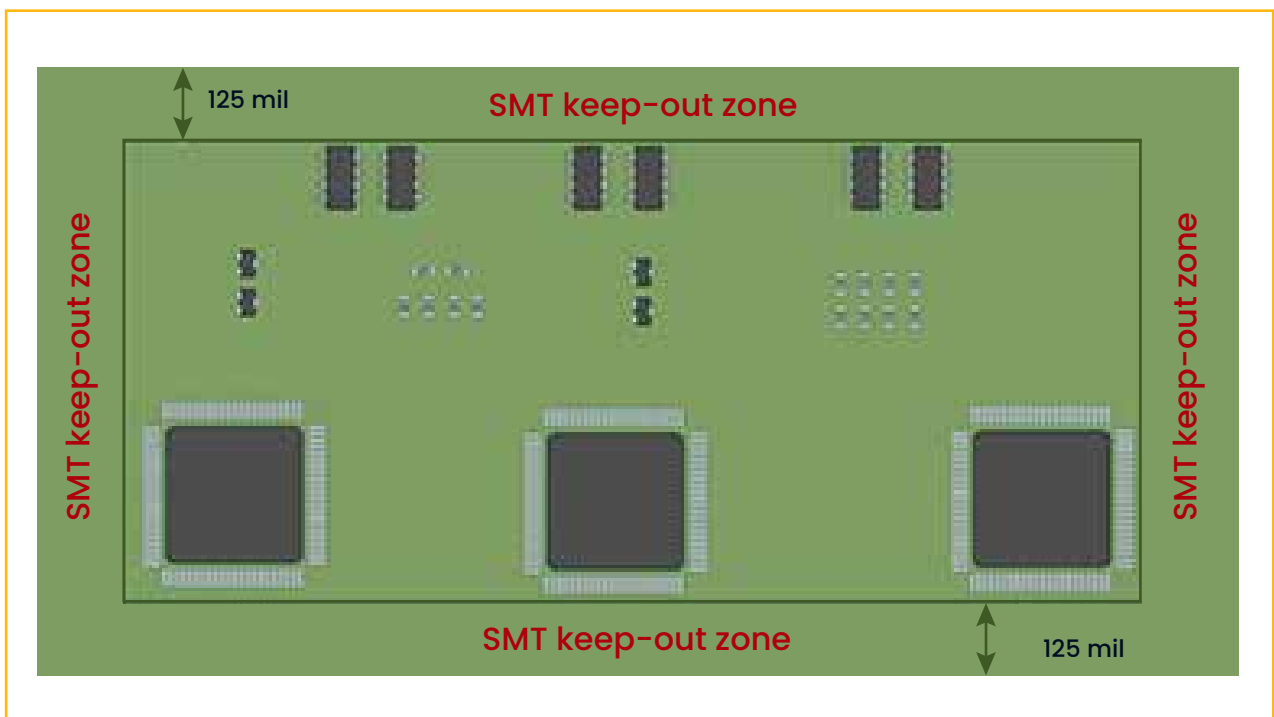
5.3 Mounting and clearance instructions

- Maintain proper component clearances to prevent tombstoning and solder bridging. Your notes should:
 - Specify minimum spacing between components as per IPC-2221 or specific assembly requirements to prevent soldering defects.
 - Define a minimum placement courtyard around chip components (including package body and land pattern) to ensure sufficient spacing for assembly and rework.



An illustration of a component courtyard

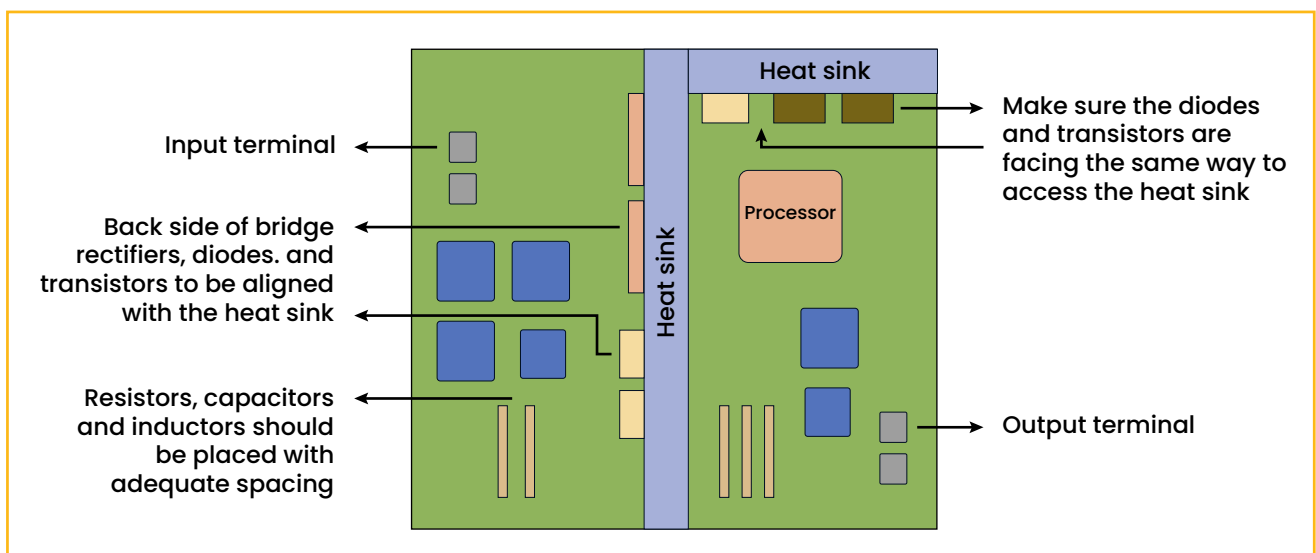
- Provide keep-out zones around large components and clearances for automated assembly equipment.



Component to board-edge clearances

- Mechanical constraints can affect part placement, especially with through-hole components, large inductors, connectors, and shields. Clearly state:
 - Maximum component height limits (especially for sub-assemblies).
 - Keep-out areas for nearby parts.
 - Clearance from edge connectors or panel break lines.
 - Required adhesive or mechanical retention methods (e.g., epoxy, press-fit).
 - No mount zones around connectors, switches, and panel cutouts.
 - If any components must be mounted after testing or post-processing.
 - Dimensional drawings or callouts for complex mounting arrangements.
 - Instructions for attaching heat sinks, thermal pads, or other thermal management devices.
 - Mechanical hardware instructions like torque specs, washer usage, adhesives, gluing, staking, and post-assembly handling.

Here's an example of an assembly drawing.



Assembly drawing with heat sink specifications

Example note:

- 0.25 mm minimum clearance for most components unless otherwise specified.
- 1.27 mm (50 mil) place bound/courtyard clearance for connectors as per standard footprints.
- 1.0 mm clearance is required for BGA devices to facilitate inspection and rework.
- Maintain a 1.5 mm clearance between C17 and heatsink U3. Maximum component height in this region shall not exceed 10 mm.
- Ensure a minimum clearance of 50 mil from the board edge for all surface mount components. All mounting holes must have a diameter tolerance of ± 2 mil.

5.4 Soldering requirements

Follow these instructions when you're writing notes for soldering requirements:

- Define the expected soldering quality and process standards. Refer to industry-recognized standards such as:
 - J-STD-001: For soldering process quality and material criteria
 - J-STD-003: For solderability testing requirements (component leads and pads)
- State the required solder alloy (e.g., SnAgCu for lead-free, SnPb for leaded).
- Specify cleaning requirements (no-clean, water-soluble, or rosin-based flux).
- Mention whether components are washable or non-washable to guide post-solder cleaning processes. Clearly mark any no-clean or sensitive components on the BOM and assembly drawings.
- Outline any unique reflow profiles essential for sensitive components or complex assemblies by providing guidelines or referencing a separate document.
- Detail any wave soldering parameters or areas to be masked for through-hole components.
- Indicate any components that require hand soldering and provide specific instructions.

Example note:

- All soldering shall conform to IPC-J-STD-001 Class 2 unless otherwise noted.
- Solderability of all parts must meet J-STD-003 prior to placement.
- Manual soldering allowed for PTH components only, using a temperature-controlled iron.
- All BGAs must undergo post-reflow X-ray inspection. No-clean flux shall be used unless otherwise specified
- U4, U6, and U9 are non-washable components. Avoid aqueous or ultrasonic cleaning in these areas.

5.5 Environmental and ROHS/REACH compliance

If your board must meet environmental or regulatory compliance, include statements that clearly indicate:

- Whether the assembly must be RoHS-compliant.
- Any substances restricted under REACH.
- Acceptable solder alloy types (e.g., SAC305).
- Any exceptions permitted (e.g., military exemptions for leaded solder).

Example note:

Assembly must comply with RoHS 3 (Directive 2015/863). No intentionally added lead or halogens. Use lead-free solder SAC305.

5.6 Notes for lead-free or high-temperature assemblies

Lead-free and high-temperature assemblies require specialized handling due to higher reflow temperatures and material limitations. Clear documentation ensures that the board survives thermal stress without delamination or warping. Your note should specify:

- Maximum reflow cycles allowed (e.g., no more than 2 reflow passes).
- Specific solder types if required (e.g., use SnAgCu for all joints).
- Bake-out instructions for moisture-sensitive parts.

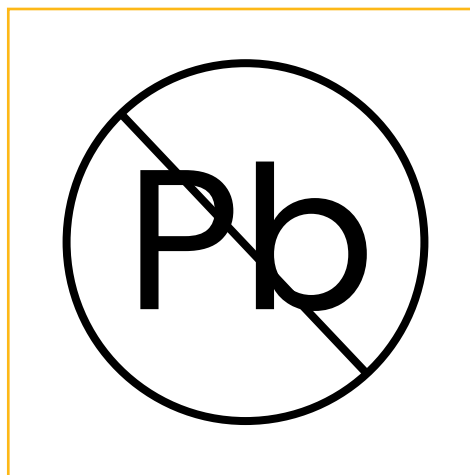
- To avoid mixing leaded and lead-free components unless explicitly allowed.
- If the PCB materials or components have specific temperature sensitivities, the assembly process must remain within safe thermal limits.

Example note:

- This board requires lead-free assembly with a maximum reflow temperature of 260°C. All components must withstand at least three reflow cycles. Moisture-sensitive devices (MSL 3+) must be baked for 24 hours before assembly.
- The IPC-1066 standard provides clear guidelines for identifying and labeling lead-free assemblies, components, and devices, ensuring consistency across global supply chains and aiding in recycling efforts.

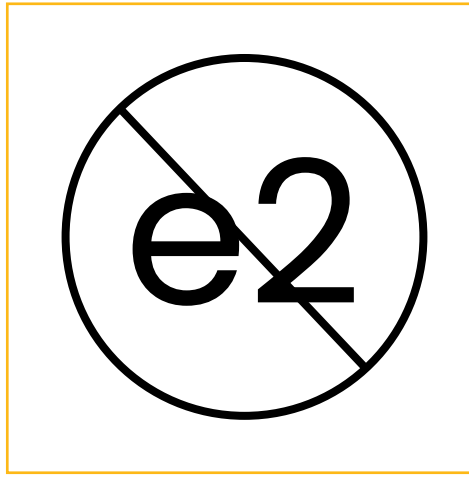
To comply with IPC-1066, include the following in your documentation:

- Use the Pb-free symbol or the phrase to identify RoHS-compliant boards with $\leq 0.1\%$ lead content.



Pb-free symbol

- Specify solder category codes (e.g., e1 = SnAgCu, e2 = SnCu) to identify the type of Pb-free solder used.



E2 solder category symbol

- Mark halogen-free boards (HF), if the base resin and reinforcement matrix meet the limits defined in IEC 61249-2-21 (≤ 1500 ppm total halogens, < 900 ppm of bromine and chlorine each).
- Position labels on PCB Layer 1 (top side), preferably at the lower right corner. Text and symbols must be legible without magnification.
- If more than one Pb-free solder alloy is used (e.g., different for reflow and wave), mention the reflow categories.
- If lead-based solder is used during repair or rework, original Pb-free category markings must be permanently obscured.

Example note:

This assembly shall comply with IPC-1066 marking standards. Apply the Pb-Free identification label per Category e1 (SnAgCu) on the top silkscreen layer. Ensure visibility and accessibility for inspection and recycling.

5.7 Testing requirements

In addition to standard electrical and functional tests, some PCBAs may require specialized post-assembly testing to verify compliance with the specifications.

In your assembly notes, clearly outline any additional testing requirements post-component placement and soldering, such as visual or automated optical inspection, in-circuit testing, X-ray inspection for BGAs and QFNs, and functional testing, including power-up and signal validation.

Additional details may include:

- Expected current draw during operation.
- Reference points for oscilloscope or multimeter probes.
- Test point dimensions and spacing in the assembly notes.
- Pass/fail criteria for each test type.
- Specific test points need to remain uncovered or accessible after assembly for debugging or programming.

Example note:

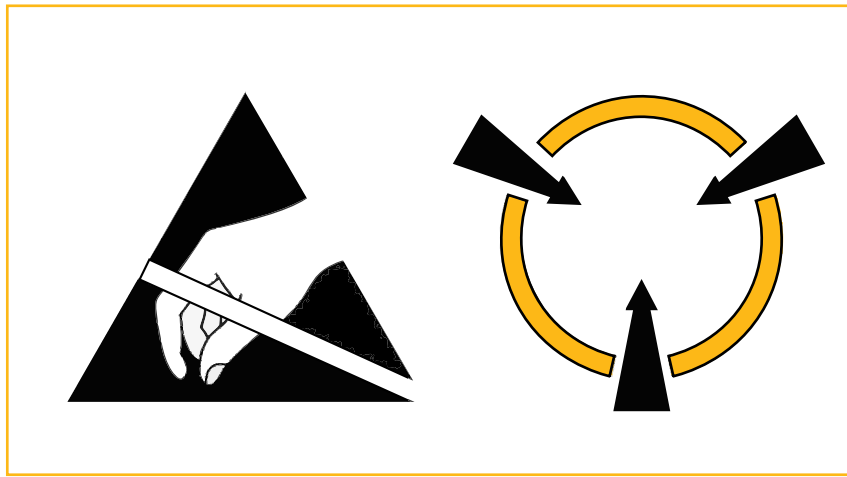
- **Perform 100% AOI on top and bottom layers. Report any tombstoned, misaligned, or cold joints for verification prior to shipment.**
- **All test points must be accessible on the top side of the board. Flying probe testing will be performed for continuity and shorts. The functional test will verify USB and Ethernet interfaces under full load conditions.**

5.8 Cautionary markings

Proper cautionary markings are essential for the safe handling and protection of your PCB, especially when it contains electrostatic discharge (ESD)-sensitive components.

Your notes should specify:

- The inclusion of appropriate ESD warning symbols in the assembly documentation.



Cautionary marking for PCB

- Ensure these symbols are clearly shown on the PCB assembly drawing, positioned near the title block or another easily seen area on the board.
- Markings can be applied using copper etching, silkscreen printing, or adhesive labels, depending on your design and manufacturing capabilities.

Example note:

This assembly includes high-voltage-sensitive devices. Apply ESD control procedures per MIL-STD-1686, Class 3, and MIL-HDBK-263 during handling and assembly.

BOM serves as a comprehensive inventory of all components required for the board, guiding purchasing, kitting, and assembly processes.

6.1. What to include in your bill of materials

An effective BOM is a structured document that includes all information necessary for purchasing, assembly, and inspection.

BOM should contain the following fields:

Field	Description
Item No.	Sequential line number
Reference designator(s)	e.g., R1, C2, U5
Quantity	Number of parts required per board
Part description	Functional info (e.g., 10kΩ ±1% resistor, 25V, 0603 package)
Manufacturer name	Full manufacturer name (e.g., Texas Instruments)
Manufacturer part number (MPN)	Exact part identifier used by the manufacturer
Footprint/package	e.g., 0603, SOIC-8, QFN-32
DNI/DNP	Mark as do not install or do not populate if applicable
Value	Component values (e.g., Resistance, capacitance, inductance)
Comments	Additional notes or special instructions

Part Number	Item	qty	Designator	Description	Mank
C2012x5R1C106K/1.25	1	1	C1	CAP CER 10UF 16V 10% X5R 0805	TDK
C2012x5R0J476M	2	1	C2	CAP CER 47UF 6.3V X5R 20% 0805	TDK
C1608x7R1A105K	3	5	C3, C132, C133, C136, C137	CAP CERAMIC 1.0UF 10V 0.1 X7R 0603	TDK
C2012X5R1A226K/1.25	4	4	C4, C16, C19, C24	CAP CER 22UF 10V 10% X5R 0805	TDK
C1005X5R0J104K	5	15	C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C20, C118, C119, C120	CAP CERAMIC .10UF 6.3V 0.1 X5R 0402	TDK

An example of a BOM

Submit your BOM in both Excel (.xls/.xlsx) and CSV formats, and ensure the design files, pick-and-place file, and BOM are synchronized in terms of reference designators and footprint names.

6.2 Handling DNI/DNP components

Not every component listed in your schematic is necessarily populated during every assembly run. Components marked as DNI (do not install) or DNP (do not populate) are included in the layout for enabling future upgrades or providing test points, but are intentionally left off the assembled board.

Item #	Quantity per board	Manufacturer Part #	Do Not Install	Reference designators	Vendor (optional)	Vendor Part Number (optional)	Value (optional)	Size / Footprint (optional)	Part description/ specs (optional, but recommended)	Manufacturer (optional)
1	2	T520L337M2R5ATE025		C12,C15	Digi-Key	399-5134-1-ND	330uF	C6032	CAP TANT 330UF 2.5V 20% SMD	Kemet
2	2	T520L337M2R5ATE025	DNI	C16,C17	Digi-Key	399-5134-1-ND	330uF	C6032	CAP TANT 330UF 2.5V 20% SMD	Kemet
3	2	RMCF0805FT2K40		R19, R24	Digi-Key	541-2.40KCCT-ND			RES SMD 2.4K OHM 1% 1/8W 0805	Vishay Dale
4	5	C3225X7R1E106M	DNI	C65,C66,C67,C68,C119	Mouser	810-C3225X7R1E106M	10uF	C1210 (3225 metric)	CAP CER 10uF 25V X7R 20% SMD 1210	TDK

An example of a BOM list

- Mark DNI/DNP parts clearly in a separate column.
- Ensure that these parts are excluded from sourcing.
- Confirm that the pick-and-place file excludes coordinates for DNI parts.
- Optionally, include a comment like “test point DNI for Rev A only” for clarity.

Avoid simply deleting DNI parts from the BOM. This may cause confusion if they still appear in the schematic or layout files.

6.3 Alternate part numbers and vendor data

Supply chain constraints or long lead times can delay your build if alternate parts aren't considered in advance. Including alternate part numbers or approved vendor options in your BOM gives your CM flexibility and improves turnaround time.

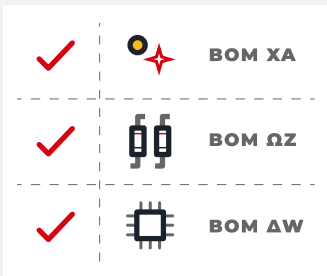
In your BOM, make sure to:

- Add a column titled approved alternates next to the primary MPN.
- Select alternates that match electrical, mechanical, and thermal specifications.
- Confirm that alternate parts are compatible with the existing footprint and orientation.
- Add a vendor name (optional but helpful) and vendor part number (VPN) for specific supply chain partners.
- Include preferred distributors (e.g., DigiKey, Mouser Electronics, Arrow Electronics).

Where compliance matters (e.g., RoHS, UL), make sure alternates also meet regulatory requirements.

A well-prepared BOM can reduce turnaround time, prevent sourcing mistakes, and ensure alignment between design and production.

Use Sierra Circuits' BOM Checker to identify issues in your material list instantly.



PCB DESIGN TOOL

BOM Checker



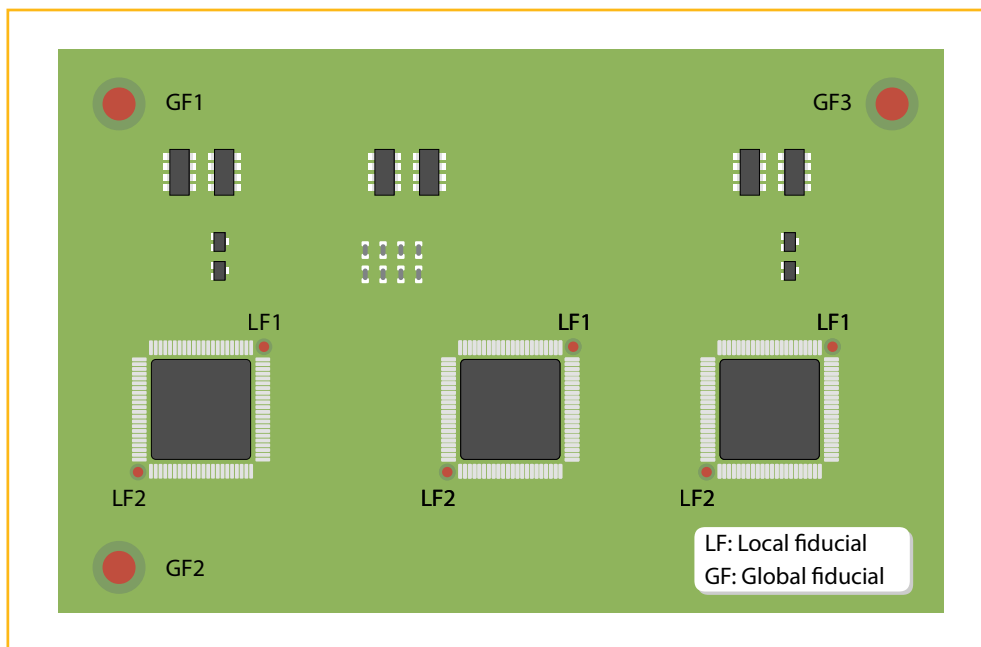
TRY TOOL

Files like BOM, pick-and-place, and assembly drawings are sufficient for many projects. However, advanced designs, such as high-density layouts, require additional documentation. They include:

7.1 Fiducials

Fiducials are copper markers used by optical vision systems to align the board or panel in X-Y space. You should include at least three global fiducials per board/panel and 2 local fiducials (diagonally) near fine-pitch or BGA components.

Specify their exact location (X-Y coordinates), diameter, and any required solder mask clearance in your assembly drawing. Ensure they are free from silkscreen or other obstructions.



Global and local fiducials

Fiducials should be:

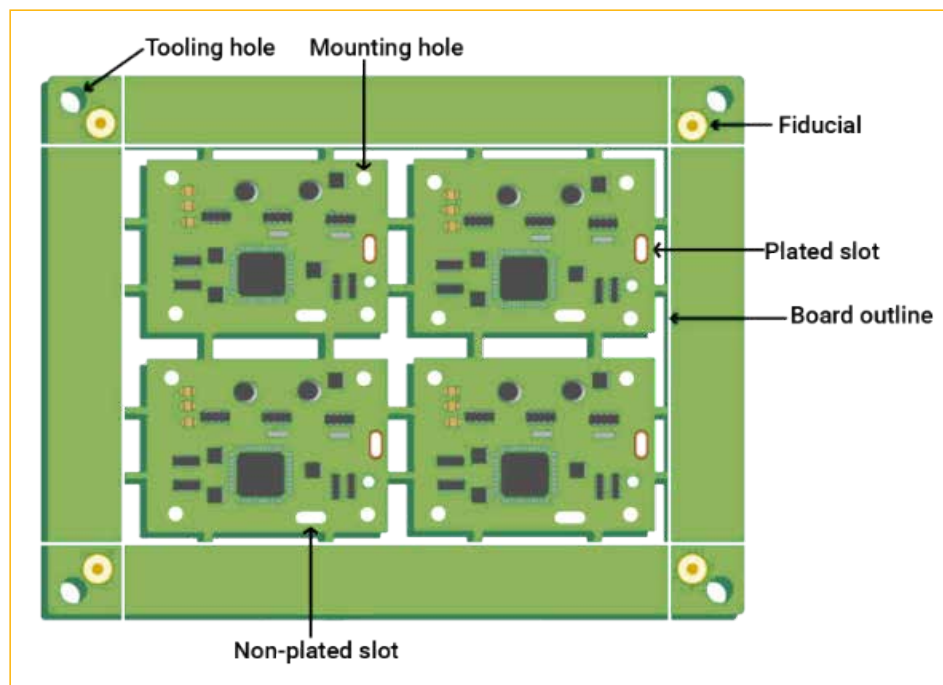
- 1.0 mm (40 mil) bare copper circles with 2.0 mm (80 mil) clearance.
- Located outside component zones. Avoid placing them too close to board edges or components.

Example note:

Include three fiducials in a triangular pattern on the board: one near the top-left corner, one near the bottom-right, and one near the center. Local fiducials must be placed adjacent to all BGA packages.

7.2 Tooling holes

Tooling holes are non-plated mechanical holes used for precise mechanical alignment and registration of a PCB during various manufacturing and assembly stages (e.g., in fixtures, test jigs, or panelization frames).



An illustration of a PCB panel with tooling holes

- Define their exact diameter, location, and whether they are plated or non-plated.
Often, two or three tooling holes are used to prevent rotation.
- Specify any keep-out areas around tooling holes to ensure unobstructed access for pins or fixtures.

Example note:

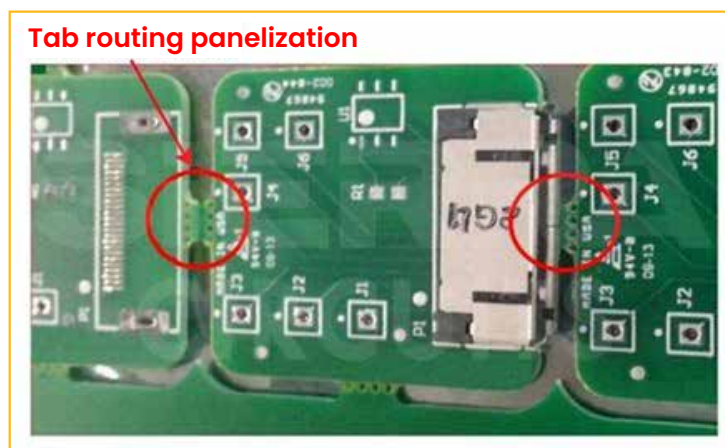
- Provide two non-plated tooling holes, 3.2 mm in diameter, located 5 mm from the X and Y edges. Minimum clearance of 3 mm must be maintained around each hole.
- Tooling holes should not be plated unless required for grounding purposes.

7.3 Alignment marks

Alignment marks are additional visual cues used for layer registration in multilayer, flexible, and rigid-flex designs. Including these marks in assembly drawings supports accurate lamination and assembly. These help reduce placement error, especially for connectors and complex modules.

7.4. Panelization instructions and breakaway tabs

Panelization groups multiple PCBs into a single array for SMT processes and should be defined in your assembly documentation.



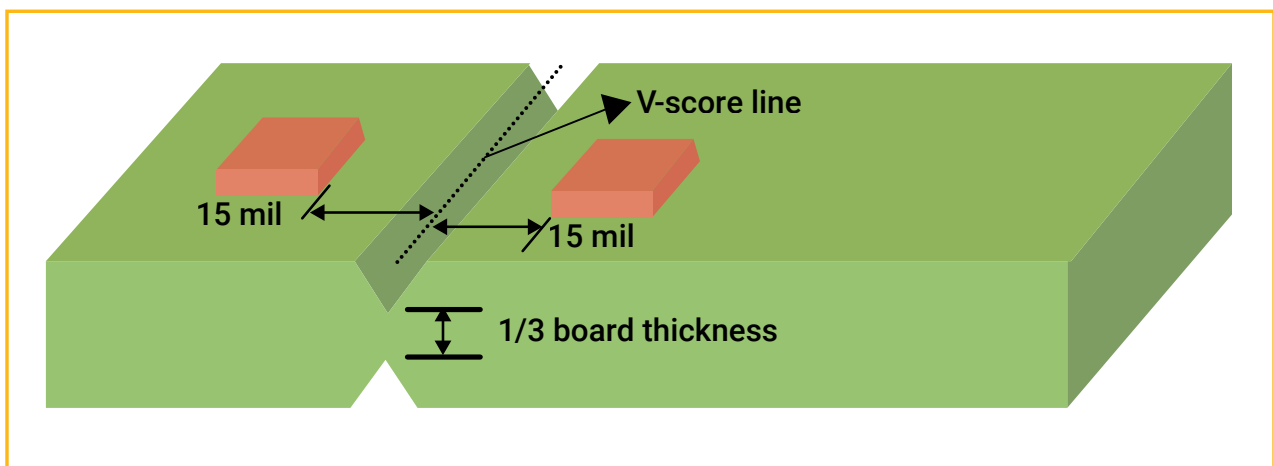
Tab routing panelization

Panelization details to be added in your assembly notes:

- Specify how an individual board should be arranged within a larger panel. This includes the number of units in the X and Y dimensions (e.g., 2x3 array).
- Indicate the preferred orientation within the panel (e.g., long edge along the X-axis).
- Define the spacing (routing channels or mouse bites) between individual units within the panel.

Method of separation:

- If V-scoring is the preferred method for separating boards, specify the depth and angle of the V-grooves. V-scoring partially cuts the board from both top and bottom, allowing easy breaking.



V-scoring on PCB panels

- If routing is used to separate boards, specify the width of the route and the placement, width, and number of breakaway tabs (mouse bites) that hold the individual boards to the panel frame.
- For complex shapes, a combination of V-scoring and routing may be used, requiring clear instructions for both.

Specify the width of the unpopulated frame around the PCB array, which is used for handling, tooling holes, and fiducials for the entire panel.

Include a panel drawing with outlines, fiducials, tooling holes, and routing paths. Detail the location and specifications of tooling holes and global fiducials placed on the panel frame for machine alignment during assembly.

Example note:

Panelize with 3 mm tabs and mouse bites between boards. Add a 10 mm border with global fiducials on each corner.

7.5 Specialized assembly instructions

Some PCBs require additional or specialized post-assembly processes or structural reinforcements beyond standard SMT and through-hole assembly. These instructions include:

7.5.1 Conformal coating

It is a protective polymeric film applied to the assembled PCB to guard against moisture, dust, chemicals, and extreme temperatures.

- If conformal coating is required, specify the type of coating material (e.g., acrylic, silicone, polyurethane), its thickness, application method (e.g., spray, dip), and curing method (UV, oven, air dry)
 - Label conformal coating as per IPC-CC-830:
 - ER – Epoxy resin
 - UR – Urethane resin
 - AR – Acrylic resin
 - SR – Silicone resin
 - XY – Paraxylylene
- Clearly define keep-out areas that must not be coated (e.g., connectors, test points, heat sinks, potentiometers, specific component tops) using clear markings or instructions on the assembly drawing.

Example note:

Apply acrylic conformal coating to the top side only. Do not coat connectors J1 and J2. Use UV-curable coating per IPC-CC-830.

7.5.2 Underfill

Underfill material is a polymeric epoxy dispensed under certain components, particularly BGAs, CSPs, and flip-chips, to enhance mechanical coupling between the component and the PCB. This improves solder joint reliability against thermal cycling and mechanical stress.

- Indicate which components require underfill, the specific type of underfill material, and any curing time and temperature details. Provide clear instructions for dispensing and specify no-flow areas.

Example note:

Dispense underfill beneath U5 (BGA). Use Lord 7556-1 or equivalent.

Cure at 150°C for 30 minutes.

7.5.3 Stiffeners

Stiffeners (e.g., FR-4, polyimide, aluminum) are rigid pieces attached to flexible circuits to provide mechanical support, usually to support components, connectors, or areas that require rigidity for handling or mounting.

- For flex or rigid-flex designs, specify the material type, thickness, and precise location of all stiffeners. Detail the adhesive type and curing process for attaching them.
- Provide clear alignment instructions for stiffener placement relative to the flex circuit.

Example note:

Apply a 0.8 mm FR4 stiffener on the bottom side under connector J3.

Attach with pressure-sensitive adhesive (3M467).

Best practices for writing effective fab notes and assembly notes

08

Your PCB design might be exceptional, but if your fabrication and assembly documentation is unclear or incomplete, it can still fail in production.

Manufacturers rely on your notes to interpret design intent. Ambiguities, contradictions, or omissions can result in delays, incorrect builds, and unnecessary back-and-forth communication that increase cost and lead time.

Here are some of the best practices for both fab and assembly notes:

- Be clear, concise, and direct. Avoid vague terms like high quality or standard process. Use measurable language (e.g., soldermask thickness: 0.8 mil minimum).
- Use standard formatting. Organize notes as a numbered or bulleted list. Use section headers to separate topics (e.g., material requirements, drilling instructions). Always use clear and consistent file naming conventions.
- Avoid redundant or conflicting instructions. Contradictions between notes, drawings, or design files lead to uncertainty and delays.
- Align notes with manufacturer capabilities. Don't request tolerances or processes that your fabricator or CM doesn't support. Use DFM/DFT feedback loops to confirm what's feasible.
- Reference industry standards. Use IPC standards (e.g., IPC-6012, IPC-A-610, J-STD-001) wherever applicable. This eliminates subjectivity and aligns your expectations with tested guidelines.

- Update version and revision history. Always include a drawing or document revision number and a change log. This helps prevent outdated builds and reduces confusion during ECOs (engineering change orders).
- Follow a logical flow, ideally mirroring the manufacturing process. Group related notes together (e.g., all drill-related notes, all solder mask notes).
- Maintain uniform font styles, sizes, and heading hierarchies throughout the document. Consistent use of bolding, italics, or indentation can improve readability.
- Use graphical indicators on assembly drawings for polarized components, and then reinforce this with notes if specific details are needed beyond the standard markings.
- Conduct a peer review of notes and drawings to catch inconsistencies or omissions.
- Use checklists to verify inclusion of critical elements (e.g., stack-up, impedance, testing requirements).

By adhering to these best practices, you can create documentation that not only facilitates manufacturing but also builds stronger, more reliable relationships with your fabrication and assembly partners.

Other resources on fab and assembly notes



Articles

- What Should a PCB Fab Drawing Include?

Knowledge Base:

- PCB Fabrication Notes
- PCB Assembly Notes
- PCB Via Tenting Design Rules and Fabrication Notes

Webinar:

- Design Documentation and How to Communicate Design Intent

NOTES: UNLESS OTHERWISE SPECIFIED

S.NO.	Particulars	
1	Fabricate per IPC	IPC-A-600D
2	PCB class	2
3	Board material	FR-370HR
4	Board finish	ENIG
5	No. of layers	4
6	Solder mask	Green
7	Silkscreen	White
8	Cut-out in Design	YES
9	Blind & buried vias	No
10	Pad on vias	No
11	Via filling	No (Non-conductive)
12	Impedance traces	Yes (See impedance note)
13	Finish: Plated through hole diameter as specified are after plating plus or minus .003 inches. The thickness of the copper plating on the wall of the hole shall be no less than .0008 inch.	
14	Fabricator to clip silkscreens using solder mask layers.	
15	FINISHED PCB SHALL MEET UL FLAMMABILITY RATING OF 94 V-0 AND BE PERMANENTLY MARKED WITH REQUIRED U.L. CODE.	
16	U.L. TEMPERATURE INDEX MINIMUM: 170 DEGREES CENTIGRADE	
17	VENDOR MUST BE LISTED IN THE "UL RECOGNIZED COMPONENT DIRECTORY".	
18	Remove all burrs and sharp edges.	

An example of standard fab notes

About Sierra Circuits

SIERRA
CIRCUITS

Sierra Circuits has been serving PCB designers and engineers with the latest technologies since 1986 and has worked with over 20,000 customers since then. We specialize in manufacturing, assembly, and HDI technology. We handle all aspects of circuit board production.

We provide our customers with DFM and DFA support, which helps produce unprecedented board quality with high reliability. By choosing us, you can eliminate miscommunication between multiple vendors and delays since we provide a single point of support.

Sierra Circuits helps PCB designers plan it right! Our engineering staff has been trained on controlled impedance and can analyze the design from a holistic point of view. Our engineering support and our stack-up team provide valuable suggestions with their knowledge of controlled impedance for high-speed designs, analog/digital, high-density board manufacturing design rules, and design for assembly guidelines. Services include system-level design, schematic capture, circuit board layout, and PCB/PCBA DFM.



Our facility

Why Sierra Circuits?

Customer-oriented approach: We collaborate closely with you and tailor solutions to meet your unique design requirements.

Rapid high-tech innovation: We work in a fast-paced environment and implement cutting-edge technologies to bring your design ideas to life.

Integrated systems: Integrated turnkey world-class team that includes design, engineering, manufacturing, support services, and supply chain that engages from the beginning of all of our partnerships.

Focus ready: We focus on complex products for defense and aerospace that require high reliability and repeatability to support mission-critical systems on time.

Sierra Circuits’ PCB manufacturing capabilities

Rigid PCBs

PCB highlights	Fabrication highlights	Quality management systems
Up to 30 layers	Routed arrays	AS9100D
Board thickness: .005" - .250"	Edge plating	ISO 9001:2015
Max panel size: 21" x 29"	Bevels	ISO 13485:2016
Min trace and space: .002"	Heat sinks	
Solder mask feature tolerance: .001"		
Blind and buried vias		

HDI and microelectronics

High tech in small form	Features	Certified for your industry
Special stack-up help / NPI review	Down to 1.5-mil trace space, 2-mil holes	Mil-spec
Up to 30 layers	Blind vias, buried vias, and other microvia techniques	Automotive
Files evaluated by engineers	Up to 5 sequential lamination cycles	Aerospace and defense
Quote sent by email within a few hours	Laser direct imaging	Medical devices
	via-in-pad technology	

Flex and rigid-flex PCBs

Surface finishes	Standard flex materials	Fabrication highlights	PCB classification
HASL	0.5 mil to 5 mil thick polyimide material	Routed array	IPC 6012, Class 1, 2, and 3
Lead-free HASL		V score	
OSP (Shikoku F2 and Entek)	1 mil to 5 mil thick copper clad base material	Edge plating	ISO:9001:2015
ENIG	Flame retardant laminate	Edge castellation	ISO 13485: 2015
Immersion silver	UL and RoHS compliant	Countersink	
Immersion white tin		Bevel	
Tin nickel		Milling	
Selective gold			

Aerospace and defense PCBs

Capabilities	Documents we provide	Ideal for
Layer count: Up to 30	Certificate of conformance	DOD product designers developing systems for mission critical functions
Minimum board thickness:10 mil	Material specifications	
Thickness tolerance: 10%	Reflow profile copy (included with first article)	Contract manufacturers who need fast, predictable lead times
Minimum trace and space width: 3 mil	Photo requirements	
Minimum drill-to-copper clearance: 6 mil	First article inspection report	Purchasing teams who require pricing and supply-chain transparency
Maximum panel size: 21" x 29"	Record of calibrated tools used during manufacturing	
Controlled impedance	FPT, AOI, ionic cleanliness report	Aerospace engineers looking to shore-up their supply chain troubles
Stacked vias		

RF and microwave PCBs

Capabilities	Certifications
Operating frequency: 10 MHz to 30 GHz	PC Manufacturers Qualification Profile (MQP)
Sodium etch hole treatment process	IPC-A-600
Full-body gold plating up to 120u"	IPC-6012 standards
Soft gold without nickel or ENIG surface finishes for enhanced board protection	ISO 9001:2015
	NADCAP
	ITAR registered
Design assistance	RoHS/REACH standards

Sierra Circuits' PCB assembly services

We offer a diverse range of PCB assembly services tailored to meet your unique requirements. Whether you need a full turnkey assembly, consigned or partially consigned assembly, we've got you covered.

Full turnkey PCB	Partially consigned assembly	Consigned assembly	Assembly capabilities
Design review	Some parts provided by the designer	Components are provided by the PCB designer	BGA, micro-BGA, QFN,CSP, lead less devices up to 0.35mm pitch, press fit components
Bare board fabrication	We will source the rest	Sierra Circuits will fully assemble your PCBs and ship them to you.	DFA
Component procurement	Supply chain resilient	Quick turn times	RoHS, leaded, indium, clean and no clean chemistry
Testing	Quick turnaround time	Fully transparent pricing	Component sizes: 0201, 01005, 08004
Quick turnaround time			Paste in-hole

To help you overcome the challenges involved in component procurement, Sierra Circuits has introduced customer-owned inventory service (COIN). It's a personalized component sourcing and stocking program. We procure and store parts for your PCBAs. This can help you significantly reduce your product development time.

Sierra Circuits' PCB design services

Whether you're beginning with a new design or enhancing an existing one, our design experts provide the assistance you need. Our PCB design services include stack-up design, signal integrity analysis, circuit board layout optimization, schematic capture, and reverse engineering.

Our capabilities include:

- High-speed digital designs, analog designs, mixed designs, power designs, and RF designs
- Controlled impedance with a tolerance of $\pm 5\%$
- Mechanical-constrained boards and enclosures
- Flex, rigid, and rigid-flex designs
- PCB designs for mobile phones
- HDI designs with via-in-pad, blind via, buried via, microvias, via stacking, and staggering technology
- Fine-pitch BGAs of 0.4 mm and 0.5 mm
- Designs with high-pin count full matrix BGAs 0.4 mm pitch, PoP package, with 2 mil/ 2 mil tracks/spacings
- Design for ROHS compliance
- Optimum impedance calculation for single-ended, differential, and coplanar-waveguide models

Advanced PCB design tools

From material selection to impedance calculation, streamline your design process by using our comprehensive suite of cutting-edge PCB design tools. These web applications enable you to expedite your design process and make the right decisions to ensure signal integrity and reliability in your circuit board.

Sierra Circuits PCB certification and registration

IPC certified for quality standards in PCB fabrication and assembly

- IPC Manufacturers Qualification Profile (MQP) for Sierra Circuits, Sunnyvale, California
- IPC-A-600 inspectors

ISO certified for military, aerospace, and medical device PCB fabrication and assembly

- ISO 9001:2015 certificate
- AS9100D - military and aerospace
- ISO 13485:2016 certificate - medical devices

Mil-Spec certified and DLA approved

- Acc 2020 CVI VQE 035113
- JCP certification approval letter

ITAR Registered

- ITAR registration letter

Underwriters Laboratory certified

- Flex and rigid flex PCBs
- Rigid PCBs

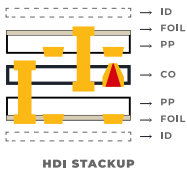
Exceptional material compliance and reporting

- RoHS / REACH compliance material composition declaration
- Conflict minerals reporting template

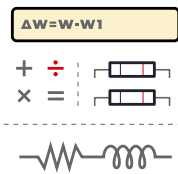
Certified minority supplier by NMSDC

- Minority business certification

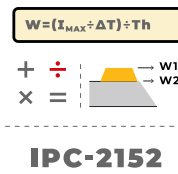
The Designer's Toolkit



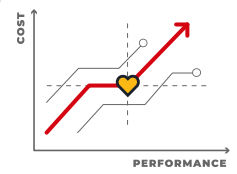
Stackup Designer



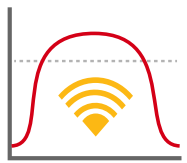
Impedance Calculator



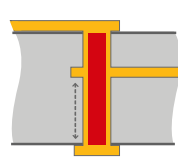
Trace Width,
Current Capacity and
Temperature Rise
Calculator



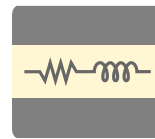
Material Selector



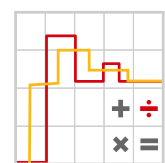
Bandwidth, Rise Time
and Critical Length
Calculator



Maximum Via
Stub Length
Calculator



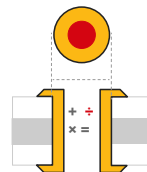
Signal Layer Estimator



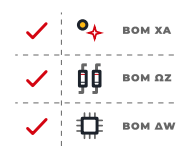
Transmission Line
Reflection Calculator



RLC Resonant
Frequency &
Impedance Calculator



Via Current
Capacity and
Temperature Rise
Calculator



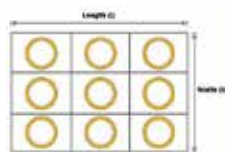
BOM Checker



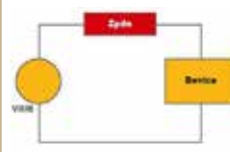
Better DFM



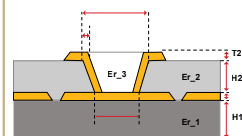
Conductor Spacing
and Voltage Calculator



Via Thermal
Resistance Calculator



Power Distribution
Network Analyzer



Via Impedance
Calculator

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